

BEC304 - NETWORK ANALYSIS

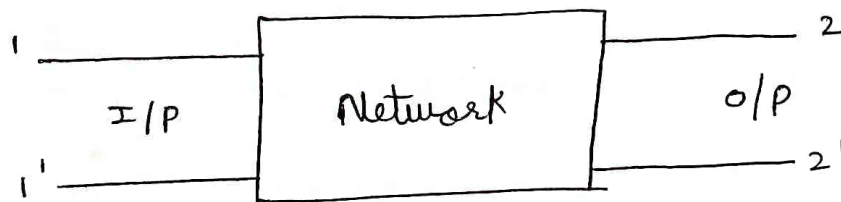
MODULE-5 TWO PORT NETWORK & RESONANCE

Introduction

A network having 2 pairs of terminals forms an important building block in many system communication, control systems. The pair of terminals at which the electrical energy enters is known as I/P terminal, while a pair of terminal which signal leaves is known as o/p terminal.

A pair of terminals at which a signal may enter or leave a n/w is called port.

A n/w having 2 pairs of terminals is known as 2 port network.



where, $1 \& 1'$ are I/P terminals
 $2 \& 2'$ are o/p terminals

A two port n/w has 4 variables $V_1, V_2, I_1 \& I_2$ out of these four, two variables can be dependent variable & other two variables are independent.

Depending on which variables are chosen as independent variables 4 sets of parameter are defined for a 2 port network they are.

① open ckt impedance (Z -parameters)

Dependent variables = V_1, V_2

independent variables = I_1, I_2

② short ckt ~~impedance~~ admittance (Y -Parameters)

Dependent variables = I_1, I_2

independent variables = V_1, V_2

③ Hybrid or h -parameter

Dependent variable = V_1, I_2

independent variable = V_2, I_1

④ Transmission / chain / ABCD / General circuit parameters

Dependent variables: V_1, I_1

independent variables: V_2, I_2

→ open ckt impedance or Z parameters:

let V_1 & V_2 be the dependent variables, I_1 & I_2 are independent variables

$$\therefore V_1 = f_1(I_1, I_2)$$

$$V_2 = f_2(I_1, I_2)$$

The i/p & o/p voltages V_1 & V_2 can be expressed in terms of i/p & o/p currents I_1 & I_2 as

$$V_1 = Z_{11}I_1 + Z_{12}I_2 \dots \textcircled{1}$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2 \dots \textcircled{2}$$

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

Put $I_2 = 0$ in eq ① & ②

$$V_1 = Z_{11} I_1$$

$$Z_{11} = \frac{V_1}{I_1}$$

$$I_2 = 0$$

It is called as i/p impedance (ohm)

$$V_2 = Z_{21} I_1$$

$$Z_{21} = \frac{V_2}{I_1}$$

$$I_2 = 0$$

It is called as forward transfer impedance

Put $I_1 = 0$ in ① & ②

$$V_1 = Z_{12} I_2$$

$$Z_{12} = \frac{V_1}{I_2}$$

$$I_1 = 0$$

It is called reverse transfer impedance

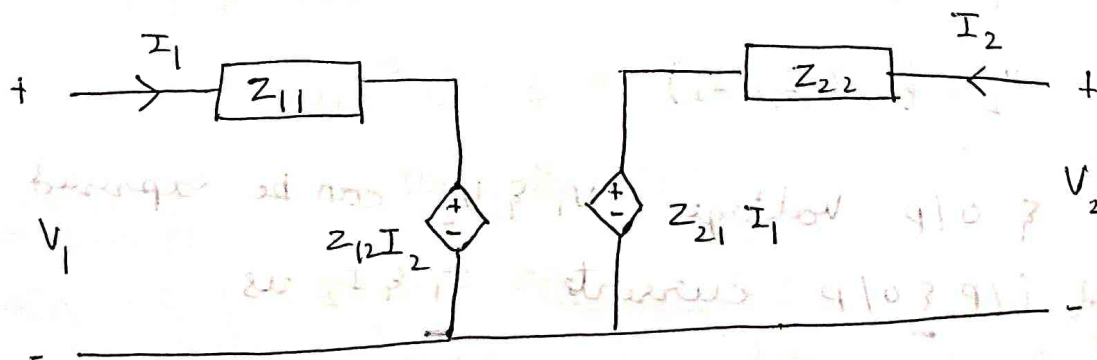
$$V_2 = Z_{22} I_2$$

$$Z_{22} = \frac{V_2}{I_2}$$

$$I_1 = 0$$

It is called as o/p impedance (ohm)

• equivalent ckt: z parameters.



→ short ckt admittance (Y Parameter)

$$I_1 = b_1 (V_1, V_2)$$

$$I_2 = b_2 (V_1, V_2)$$

$$I_1 = y_{11} V_1 + y_{12} V_2 \quad \text{--- (1)}$$

$$I_2 = y_{21} V_1 + y_{22} V_2 \quad \text{--- (2)}$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

put $V_2 = 0$ in eq (1) & (2)

$$I_1 = y_{11} V_1$$

$$\boxed{y_{11} = \frac{I_1}{V_1}}$$

$$V_2 = 0$$

Input admittance (mho)

$$I_2 = y_{21} V_1$$

$$\boxed{y_{21} = \frac{I_2}{V_1}}$$

Forward transfer admittance

Put $V_1 = 0$ in eq (1) & (2)

$$\boxed{y_{12} = \frac{I_1}{V_2}}$$

$$V_1 = 0$$

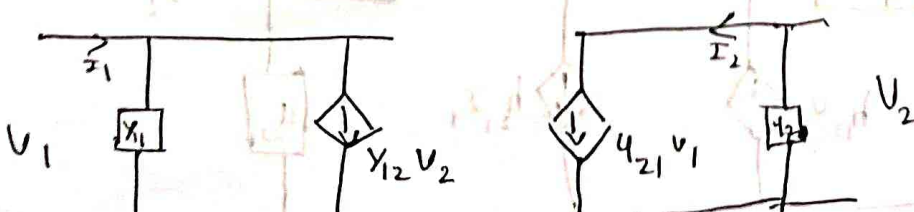
Reverse transfer admittance

$$\boxed{y_{22} = \frac{I_2}{V_2}}$$

$$V_1 = 0$$

Output admittance (mho)

• equivalent Y parameter model



→ H - Parameter :

$$V_1 = b_1 (I_1, V_2)$$

$$I_2 = b_2 (I_1, V_2)$$

$$V_1 = H_{11} I_1 + H_{12} V_2 \quad \text{--- (1)}$$

$$I_2 = H_{21} I_1 + H_{22} V_2 \quad \text{--- (2)}$$

put $V_2 = 0$ in (1) & (2)

$$V_1 = H_{11} I_1$$

$$H_{11} = \frac{V_1}{I_1}$$

$$V_2 = 0$$

It is called as input impedance (ohm)

$$H_{21} = \frac{I_2}{I_1}$$

$$V_2 = 0$$

Forward current gain

put $I_1 = 0$ in (1) & (2)

$$H_{12} = \frac{V_1}{V_2}$$

$$I_1 = 0$$

Reverse voltage gain

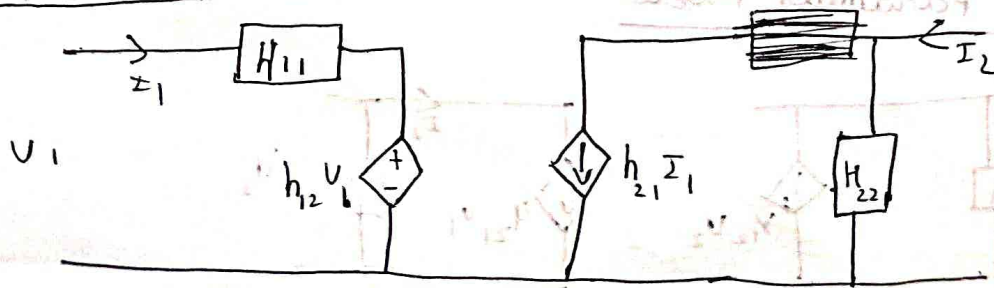
$$H_{22} = \frac{I_2}{V_2}$$

$$I_1 = 0$$

It is called as output admittance (mho)

→ It contains different parameters such as input impedance, current gain, AV & admittance hence called Hybrid parameter

→ H parameter equivalent module:



-> ABCD Parameters:

Dependent: V_1, I_1

Independent: V_2, I_2

ABCD parameter is used in power transmission lines

$$V_1 = b_1(V_2, I_2)$$

$$I_1 = b_1(V_2, I_2)$$

$$V_1 = A V_2 - B I_2 \quad \text{--- (1)}$$

$$I_1 = C V_2 - D I_2 \quad \text{--- (2)}$$

-ve sign for I_2 is due to the fact that I_2 flows in a direction ^{opp} to that of current

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

Put $I_2 = 0$ in (1) & (2)

$$A = \frac{V_1}{V_2}$$

$$C = \frac{I_1}{V_2}$$

$$I_2 = 0$$

Put $V_2 = 0$ in (1) & (2)

$$B = -\frac{V_1}{I_1}$$

$$D = -\frac{I_1}{I_2}$$

$$V_2 = 0$$

show that $z = [y]$

We know that, $[V] = [I][z]$

Premultiply $[z]^{-1}$ on both sides

$$[V][z]^{-1} = [I]$$

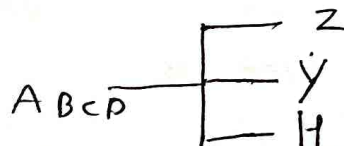
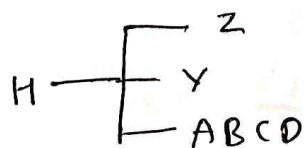
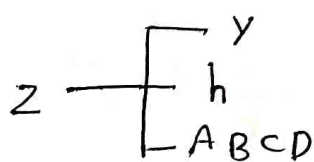
$$[V][z]^{-1} = [I]$$

$$[V][z]^{-1} = [I]$$

$$[z]^{-1} = [Y]$$

$$[Y] = [z]^{-1} = \frac{1}{[z]}$$

→ Interrelation between parameters



• Z Parameter in terms of Y parameters:

WKT, Z parameters are given by

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \dots \textcircled{1}$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \dots \textcircled{2}$$

Y parameters are given by

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \dots \textcircled{3}$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \dots \textcircled{4}$$

represent ③ & ④

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

V_1 & V_2 is obtained by cramer's rule

$$V_1 = \frac{\begin{bmatrix} I_1 & Y_{12} \\ I_2 & Y_{22} \end{bmatrix}}{\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix}} = \frac{Y_{22} I_1 - Y_{12} I_2}{Y_{11} Y_{22} - Y_{12} Y_{21}}$$

$$V_1 = \frac{Y_{22} I_1 - Y_{12} I_2}{\Delta Y}$$

$$V_1 = \frac{Y_{22} I_1}{\Delta Y} - \frac{Y_{12} I_2}{\Delta Y} \quad \dots \dots \textcircled{5} \quad Z_{11} = \frac{Y_{22}}{\Delta Y} \quad , \quad Z_{12} = -\frac{Y_{12}}{\Delta Y}$$

composing ⑤ & ①

$$V_1 = \frac{Y_{22}}{\Delta Y} I_1 + \left(-\frac{Y_{12}}{\Delta Y} \right) I_2$$

$$V_2 = \frac{\begin{bmatrix} Y_{11} & I_1 \\ Y_{21} & I_2 \end{bmatrix}}{Y_{11} Y_{22} - Y_{12} Y_{21}}$$

$$V_2 = \frac{Y_{11} I_2 - Y_{21} I_1}{\Delta Y}$$

$$V_2 = \frac{Y_{11}}{\Delta Y} I_2 - \frac{Y_{21}}{\Delta Y} I_1 \quad \dots \dots \textcircled{6}$$

composing ⑥ & ②

$$V_2 = \frac{Y_{11}}{\Delta Y} I_2 + \left(-\frac{Y_{21}}{\Delta Y} \right) I_1$$

$$Z_{21} = \frac{Y_{11}}{\Delta Y} \quad , \quad Z_{22} = -\frac{Y_{21}}{\Delta Y}$$

$$[Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{X_{22}}{\Delta Y} & -\frac{Y_{12}}{\Delta Y} \\ -\frac{Y_{21}}{\Delta Y} & \frac{X_{11}}{\Delta Y} \end{bmatrix}$$

→ Z parameter in terms of H parameters.

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad \dots \quad (1)$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad \dots \quad (2)$$

H parameter

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \dots \quad (3)$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \dots \quad (4)$$

from (4), V_2

$$V_2 = \frac{I_2 - h_{21} I_1}{h_{22}}$$

$$V_2 = \frac{1}{h_{22}} I_2 - \frac{h_{21}}{h_{22}} I_1 \quad \dots \quad (5)$$

comparing (5) & (2)

$$Z_{22} = \frac{1}{h_{22}}$$

$$Z_{21} = -\frac{h_{21}}{h_{22}}$$

subs V_2 in Eqn (3)

$$V_1 = h_{11} I_1 + h_{12} \left[\frac{1}{h_{22}} I_2 - \frac{h_{21}}{h_{22}} I_1 \right]$$

$$V_1 = h_{11} I_1 + \left[\frac{h_{12}}{h_{22}} I_2 - \frac{h_{21} h_{12}}{h_{22}} I_1 \right]$$

$$V_1 = \frac{h_{12}}{h_{22}} I_2 + \left[\frac{h_{22} h_{11} - h_{21} h_{12}}{h_{22}} \right] I_1 \dots \textcircled{6}$$

comparing $\textcircled{6}$ $\textcircled{5}$ $\textcircled{1}$

$$Z_{11} = \frac{\Delta h}{h_{22}}$$

$$Z_{12} = \frac{h_{12}}{h_{22}}$$

$$[Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{\Delta h}{h_{22}} & \frac{h_{12}}{h_{22}} \\ -\frac{h_{21}}{h_{22}} & \frac{1}{h_{22}} \end{bmatrix}$$

→ Z parameter in terms of ABCD parameter.

Z parameter

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \dots \textcircled{1}$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \dots \textcircled{2}$$

ABCD parameter

$$V_1 = A V_2 - B I_2 \dots \textcircled{3}$$

$$I_1 = C V_2 - D I_2 \dots \textcircled{4}$$

from $\textcircled{4}$, V_2

$$V_2 = \frac{I_1 + D I_2}{C}$$

$$V_2 = \frac{1}{C} I_1 + \frac{D}{C} I_2 \dots \textcircled{5}$$

comparing $\textcircled{5}$ & $\textcircled{2}$

$$Z_{21} = \frac{1}{C}$$

$$Z_{22} = \frac{D}{C}$$

subs ⑤ in ③

$$V_1 = A \left[\frac{1}{C} I_1 + \frac{D}{C} I_2 \right] - B I_2$$

$$V_1 = \frac{A}{C} I_1 + \frac{AD}{C} I_2 - B I_2$$

$$V_1 = \frac{A}{C} I_1 + \left[\frac{AD - BC}{C} \right] I_2$$

$$V_1 = \frac{A}{C} I_1 + \left[\frac{\Delta T}{C} \right] I_2 \quad \dots \quad \textcircled{6}$$

comparing ⑥ with ①

$$\boxed{Z_{11} = \frac{A}{C}}$$

$$\boxed{Z_{12} = \frac{\Delta T}{C}}$$

$$[Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{A}{C} & \frac{\Delta T}{C} \\ \frac{1}{C} & \frac{D}{C} \end{bmatrix}$$

Z-Parameter	y parameter	h parameter	ABCD Parameter
$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}$	$\begin{bmatrix} \frac{y_{22}}{\Delta y} & -\frac{y_{12}}{\Delta y} \\ -\frac{y_{21}}{\Delta y} & \frac{y_{11}}{\Delta y} \end{bmatrix}$	$\begin{bmatrix} \frac{\Delta h_{11}}{h_{22}} & \frac{h_{12}}{h_{22}} \\ -\frac{h_{21}}{h_{22}} & \frac{1}{h_{22}} \end{bmatrix}$	$\begin{bmatrix} \frac{A}{C} & \frac{\Delta T}{C} \\ \frac{1}{C} & \frac{D}{C} \end{bmatrix}$

→ y in terms of z parameters:

y parameter

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \quad \dots \quad (1)$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \quad \dots \quad (2)$$

z parameter

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad \dots \quad (3)$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad \dots \quad (4)$$

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$$I_1 = \frac{V_1 Z_{22} - V_2 Z_{12}}{\begin{vmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{vmatrix}}} = \frac{V_1 Z_{22} - V_2 Z_{12}}{\Delta Z}$$

$$I_1 = \frac{Z_{22}}{\Delta Z} V_1 - \frac{Z_{12}}{\Delta Z} V_2 \quad \dots \quad (5)$$

comparing (5) & (1)

$$Y_{11} = \frac{Z_{22}}{\Delta Z}$$

$$Y_{12} = \frac{-Z_{12}}{\Delta Z}$$

$$I_2 = \frac{\begin{vmatrix} Z_{11} & V_1 \\ Z_{21} & V_2 \end{vmatrix}}{\begin{vmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{vmatrix}}} = \frac{Z_{11} V_2 - Z_{21} V_1}{\Delta Z}$$

$$I_2 = \frac{Z_{11}}{\Delta Z} V_2 - \frac{Z_{21}}{\Delta Z} V_1 \quad \dots \quad (6)$$

comparing (6) & (2)

$$Y_{21} = \frac{-Z_{21}}{\Delta Z}$$

$$Y_{22} = \frac{Z_{11}}{\Delta Z}$$

$$\begin{bmatrix} Y \\ \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} \frac{Z_{22}}{\Delta Z} & -\frac{Z_{12}}{\Delta Z} \\ -\frac{Z_{21}}{\Delta Z} & \frac{Z_{11}}{\Delta Z} \end{bmatrix}$$

-> y parameters in terms of h parameters:

y parameters:

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \dots \textcircled{1}$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \dots \textcircled{2}$$

h parameters:

$$V_1 = h_{11} I_1 + h_{12} V_2 \dots \textcircled{3}$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \dots \textcircled{4}$$

from eqn $\textcircled{3}$,

$$I_1 = \frac{V_1 - h_{12} V_2}{h_{11}}$$

$$I_1 = \frac{1}{h_{11}} V_1 + \left(-\frac{h_{12}}{h_{11}} \right) V_2$$

$$I_1 = \frac{1}{h_{11}} V_1 - \frac{h_{12}}{h_{11}} V_2 \dots \textcircled{5}$$

comparing $\textcircled{5}$ & $\textcircled{1}$

$$\boxed{Y_{11} = \frac{1}{h_{11}}}$$

$$\boxed{Y_{12} = -\frac{h_{12}}{h_{11}}}$$

subs $\textcircled{5}$ in $\textcircled{4}$

$$I_2 = h_{21} \left(\frac{1}{h_{11}} V_1 - \frac{h_{12}}{h_{11}} V_2 \right) + h_{22} V_2$$

$$I_2 = \frac{h_{21}}{h_{11}} V_1 + \frac{h_{21} h_{12} - h_{11} h_{22}}{h_{11}} V_2 \dots \textcircled{6}$$

comparing ① & ②

$$y_{21} = \frac{h_{21}}{h_{11}}$$

$$y_{22} = \frac{\Delta h}{h_{11}}$$

$$\therefore [Y] = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} = \begin{bmatrix} \frac{1}{h_{11}} & -\frac{h_{12}}{h_{11}} \\ \frac{h_{21}}{h_{11}} & \frac{\Delta h}{h_{11}} \end{bmatrix}$$

→ y parameters in terms of abcd parameters:

y parameters :

$$I_1 = y_{11} V_1 + y_{12} V_2 \quad \dots \quad \textcircled{1}$$

$$I_2 = y_{21} V_1 + y_{22} V_2 \quad \dots \quad \textcircled{2}$$

abcd parameters :

$$V_1 = A V_2 - \cancel{B I_2} B I_2 \quad \dots \quad \textcircled{3}$$

$$I_1 = C V_2 - \cancel{D I_2} D I_2 \quad \dots \quad \textcircled{4}$$

from eqn ③

$$I_2 = -\frac{1}{B} V_1 + \frac{A}{B} V_2 \quad \dots \quad \textcircled{5}$$

comparing ⑤ & ②

$$y_{21} = -\frac{1}{B}$$

$$y_{22} = \frac{A}{B}$$

subs ⑤ in ④

$$I_1 = C V_2 - D \left(\frac{A}{B} V_2 - \frac{1}{B} V_1 \right)$$

$$I_1 = C V_2 - \frac{DA}{B} V_2 + \frac{D}{B} V_1$$

$$I_1 = \frac{D}{B} V_1 + \frac{C - DA}{B} V_2$$

$$I_1 = \frac{D}{B} V_1 + \frac{BC - DA}{B} V_2 \quad \rightarrow - (AD - BC)$$

Since $BC - DA$ this is $-(DA - BC)$ by rearranging

$$I_1 = \frac{D}{B} V_1 - \frac{\Delta T}{B} V_2 \quad \text{--- (6)}$$

comparing eqn (6) & (1)

$$Y_{11} = \frac{D}{B}$$

$$Y_{12} = -\frac{\Delta T}{B}$$

$$\therefore [Y] = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} \frac{D}{B} & -\frac{\Delta T}{B} \\ -\frac{1}{B} & \frac{A}{B} \end{bmatrix}$$

y - parameter	z - parameter	h - parameter	ABCD parameter
$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix}$	$\begin{bmatrix} \frac{Z_{22}}{\Delta Z} & -\frac{Z_{12}}{\Delta Z} \\ -\frac{Z_{21}}{\Delta Z} & \frac{Z_{11}}{\Delta Z} \end{bmatrix}$	$\begin{bmatrix} \frac{1}{h_{11}} & -\frac{h_{12}}{h_{11}} \\ \frac{h_{21}}{h_{11}} & \frac{\Delta h}{h_{11}} \end{bmatrix}$	$\begin{bmatrix} \frac{D}{B} & -\frac{\Delta T}{B} \\ -\frac{1}{B} & \frac{A}{B} \end{bmatrix}$

→ h parameter in terms of z parameters

h parameter

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \text{--- (1)}$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \text{--- (2)}$$

z parameter

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad \text{--- (3)}$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad \text{--- (4)}$$

from eqn (4)

$$I_2 = \frac{V_2 - Z_{21} I_1}{Z_{22}}$$

$$I_2 = \frac{1}{Z_{22}} V_2 - \frac{Z_{21}}{Z_{22}} I_1 \quad \dots (5)$$

comparing (5) and (2)

$$h_{22} = \frac{1}{Z_{22}}$$

$$h_{21} = -\frac{Z_{21}}{Z_{22}}$$

subs eqn (5) in (3)

$$V_1 = Z_{11} I_1 + Z_{12} \left(\frac{1}{Z_{22}} V_2 - \frac{Z_{21}}{Z_{22}} I_1 \right)$$

$$V_1 = Z_{11} I_1 + \frac{Z_{12}}{Z_{22}} V_2 - \frac{Z_{12} Z_{21}}{Z_{22}} I_1$$

$$V_1 = \frac{Z_{12}}{Z_{22}} V_2 + \frac{Z_{11} Z_{22} - Z_{12} Z_{21}}{Z_{22}} I_1$$

$$V_1 = \frac{Z_{12}}{Z_{22}} V_2 + \frac{\Delta Z}{Z_{22}} I_1 \quad \dots (6)$$

comparing (6) & (1)

$$h_{11} = \frac{\Delta Z}{Z_{22}}$$

$$h_{12} = \frac{Z_{12}}{Z_{22}}$$

$$\therefore [h] = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} \frac{\Delta Z}{Z_{22}} & \frac{Z_{12}}{Z_{22}} \\ -\frac{Z_{21}}{Z_{22}} & \frac{1}{Z_{22}} \end{bmatrix}$$

→ h parameter in terms of y parameter.

h parameter:

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \text{--- (1)}$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \text{--- (2)}$$

y parameter:

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \quad \text{--- (3)}$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \quad \text{--- (4)}$$

from eqn (3)

$$V_1 = \frac{I_1}{Y_{11}} - \frac{Y_{12}}{Y_{11}} V_2$$

$$V_1 = \frac{1}{Y_{11}} I_1 - \frac{Y_{12}}{Y_{11}} V_2 \quad \text{--- (5)}$$

comparing (5) & (1)

$$\boxed{h_{11} = \frac{1}{Y_{11}}}$$

$$\boxed{h_{12} = -\frac{Y_{12}}{Y_{11}}}$$

subs eqn (5) in (4)

$$I_2 = Y_{21} \left(\frac{1}{Y_{11}} I_1 - \frac{Y_{12}}{Y_{11}} V_2 \right) + Y_{22} V_2$$

$$I_2 = \frac{Y_{21}}{Y_{11}} I_1 - \frac{Y_{21} Y_{12}}{Y_{11}} V_2 + Y_{22} V_2$$

$$I_2 = \frac{Y_{21}}{Y_{11}} I_1 + \frac{Y_{22} Y_{11} - Y_{21} Y_{12}}{Y_{11}} V_2 \quad \text{--- (6)}$$

comparing ⑥ & ②

$$h_{21} = \frac{Y_{21}}{Y_{11}}$$

$$h_{22} = \frac{\Delta Y}{Y_{11}}$$

$$[h] = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} \frac{1}{Y_{11}} & -\frac{Y_{12}}{Y_{11}} \\ \frac{Y_{21}}{Y_{11}} & \frac{\Delta Y}{Y_{11}} \end{bmatrix}$$

→ h parameter in terms of abc d parameter:

h parameter:

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \dots \quad (1)$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \dots \quad (2)$$

abcd parameter:

$$V_1 = A V_2 - B I_2 \quad \dots \quad (3)$$

$$I_1 = C V_2 - D I_2 \quad \dots \quad (4)$$

from eqn ④

$$I_2 = \frac{C V_2 - I_1}{D}$$

$$I_2 = \frac{C}{D} V_2 - \frac{1}{D} I_1 \quad \dots \quad (5)$$

comparing ⑤ & ②

$$h_{21} = -\frac{1}{D}$$

$$h_{22} = \frac{C}{D}$$

Subs ⑤ in ③

$$V_1 = A V_2 - B \left(\frac{C}{D} V_2 - \frac{1}{D} I_1 \right)$$

$$V_1 = A V_2 - B \left(\frac{C}{D} V_2 - \frac{1}{D} I_1 \right)$$

$$V_1 = \frac{B}{D} I_1 + \left[A - \frac{BC}{D} \right] V_2$$

$$V_1 = \frac{B}{D} I_1 + \frac{DA - BC}{D} V_2$$

$$V_1 = \frac{B}{D} I_1 + \frac{\Delta T}{D} V_2 \dots \textcircled{6}$$

comparing $\textcircled{6}$ & $\textcircled{1}$

$$h_{11} = \frac{B}{D}$$

$$h_{12} = \frac{\Delta T}{D}$$

$$\therefore [h] = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} \frac{B}{D} & \frac{\Delta T}{D} \\ -\frac{1}{D} & \frac{C}{D} \end{bmatrix}$$

h Parameter	Z Parameter	Y Parameter	ABCD Parameter
$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix}$	$\begin{bmatrix} \frac{\Delta Z}{Z_{22}} & \frac{Z_{12}}{Z_{22}} \\ -\frac{Z_{21}}{Z_{22}} & \frac{1}{Z_{22}} \end{bmatrix}$	$\begin{bmatrix} \frac{1}{Y_{11}} & -\frac{Y_{12}}{Y_{11}} \\ \frac{Y_{21}}{Y_{11}} & \frac{\Delta Y}{Y_{11}} \end{bmatrix}$	$\begin{bmatrix} \frac{B}{D} & \frac{\Delta T}{D} \\ -\frac{1}{D} & \frac{C}{D} \end{bmatrix}$

-> ABCD Parameter in terms of Z parameter:

ABCD parameter

$$V_1 = A V_2 - B I_2 \dots \textcircled{1}$$

$$I_1 = C V_2 - D I_2 \dots \textcircled{2}$$

Z parameter

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \dots \textcircled{3}$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \dots \textcircled{4}$$

from (4)

$$I_1 = \frac{V_2 - Z_{22} I_2}{Z_{21}}$$

$$I_1 = -\frac{Z_{22}}{Z_{21}} I_2 + \frac{1}{Z_{21}} V_2 \quad \dots (5)$$

Subst (5) in (2)

$$D = \frac{Z_{22}}{Z_{21}}$$

$$A = \frac{1}{Z_{21}}$$

Subst I_1 in (3)

$$V_1 = Z_{11} \left(\frac{V_2}{Z_{21}} - \frac{Z_{22}}{Z_{21}} I_2 \right) - Z_{12} I_2$$

$$V_1 = \frac{Z_{11}}{Z_{21}} V_2 - \left(\frac{Z_{11} Z_{22} - Z_{12} Z_{21}}{Z_{21}} \right) I_2 \quad \dots (6)$$

comparing eqn (6) and (1)

$$A = \frac{Z_{11}}{Z_{21}}$$

$$B = \frac{\Delta Z}{Z_{21}}$$

$$\therefore \begin{bmatrix} A & B & C & D \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \frac{Z_{11}}{Z_{21}} & \frac{\Delta Z}{Z_{21}} \\ \frac{1}{Z_{21}} & \frac{Z_{22}}{Z_{21}} \end{bmatrix}$$

→ ABCD Parameters in terms of Y Parameter:

ABCD parameter:

$$V_1 = A V_2 - B I_2 \quad \dots (1)$$

$$I_1 = C V_2 - D I_2 \quad \dots (2)$$

Y Parameter:

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \quad \dots (3)$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \quad \dots (4)$$

from ④

$$V_1 = \frac{I_2 - Y_{22} V_2}{Y_{21}}$$

$$V_1 = \frac{1}{Y_{21}} I_2 - \frac{Y_{22}}{Y_{21}} V_2 \dots \textcircled{5}$$

comparing ⑤ and ①

$$A = -\frac{Y_{22}}{Y_{21}}$$

$$B = -\frac{1}{Y_{21}}$$

sub ⑤ in ③

$$I_1 = \frac{Y_{11}}{Y_{21}} I_2 - \frac{Y_{11} Y_{22}}{Y_{21}} V_2 + Y_{12} V_2$$

$$I_1 = \frac{Y_{11}}{Y_{21}} I_2 + \left(-\frac{Y_{11} Y_{22} + Y_{12} Y_{21} V_2}{Y_{21}} \right) \dots \textcircled{6}$$

comparing ⑥ & ②

$$D = -\frac{Y_{11}}{Y_{21}}$$

$$C = -\frac{\Delta Y}{Y_{21}}$$

$$\therefore \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} -\frac{Y_{22}}{Y_{21}} & -\frac{1}{Y_{21}} \\ -\frac{\Delta Y}{Y_{21}} & -\frac{Y_{11}}{Y_{21}} \end{bmatrix}$$

→ ABCD in terms of h parameters

ABCD Parameters:

$$V_1 = A V_2 - B I_2 \dots \textcircled{1}$$

$$I_1 = C V_2 - D I_2 \dots \textcircled{2}$$

h parameters:

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \dots (3)$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \dots (4)$$

from (4)

$$I_1 = \frac{I_2}{h_{21}} - \frac{h_{22}}{h_{21}} V_2 \quad \dots (5)$$

composing (5) & (2)

$$C = -\frac{h_{22}}{h_{21}}$$

$$D = -\frac{1}{h_{21}}$$

subs (5) in (3)

$$V_1 = \frac{h_{11}}{h_{21}} I_2 - \frac{h_{22} h_{11}}{h_{21}} V_2 + h_{12} V_2$$

$$V_1 = \frac{h_{11}}{h_{21}} I_2 + \left(-\frac{h_{22} h_{11} + h_{12} h_{21}}{h_{21}} \right) V_2 \quad \dots (6)$$

composing (6) & (1)

$$A = -\frac{\Delta h}{h_{21}}$$

$$B = -\frac{h_{11}}{h_{21}}$$

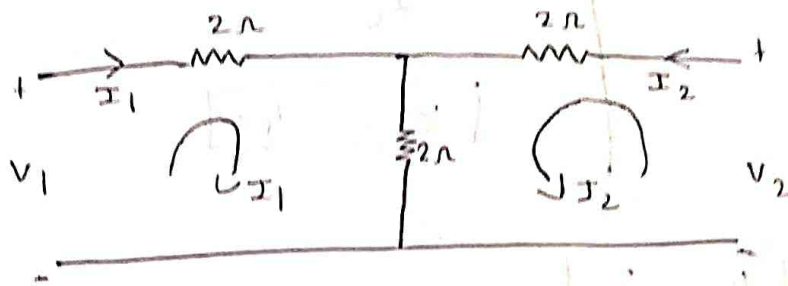
$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \frac{\Delta h}{h_{21}} & -\frac{h_{11}}{h_{21}} \\ -\frac{h_{22}}{h_{21}} & -\frac{1}{h_{21}} \end{bmatrix}$$

	Z Parameter	Y parameter	h parameters
$\begin{bmatrix} A & B \\ C & D \end{bmatrix}$	$\begin{bmatrix} \frac{Z_{11}}{Z_{21}} & \frac{\Delta Z}{Z_{21}} \\ \frac{1}{Z_{21}} & \frac{Z_{22}}{Z_{21}} \end{bmatrix}$	$\begin{bmatrix} -\frac{Y_{22}}{Y_{21}} & -\frac{1}{Y_{21}} \\ -\frac{\Delta Y}{Y_{21}} & -\frac{Y_{11}}{Y_{21}} \end{bmatrix}$	$\begin{bmatrix} -\frac{\Delta h}{h_{21}} & -\frac{h_{11}}{h_{21}} \\ -\frac{h_{22}}{h_{21}} & -\frac{1}{h_{21}} \end{bmatrix}$

→ Problem 3:

1) Find Y & Z parameter for the circuit shown

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Soln: Z parameter are given by:

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

Y parameter are given by:

$$I_1 = Y_{11} V_1 + Y_{12} V_2$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2$$

apply KVL to loop 1

$$-2 I_1 - 2(I_1 + I_2) + V_1 = 0$$

$$-V_1 = -4 I_1 - 2 I_2 \quad \text{or} \quad V_1 = 4 I_1 + 2 I_2 \quad \text{--- (1)}$$

apply KVL to loop 2

$$-2 I_2 - 2(I_2 + I_1) + V_2 = 0$$

$$V_2 = 4 I_2 + 2 I_1 \quad \text{--- (2)}$$

sub $I_2 = 0$ in eqn (1) & (2)

$$\frac{V_1}{I_1} = 4 = Z_{11}$$

$$\therefore Z_{11} = 4$$

$$\frac{V_2}{I_1} = 2 = Z_{21}$$

$$\therefore Z_{21} = 2$$

put $I_1 = 0$ in ① & ②

$$\frac{V_1}{I_2} = 2$$

$$Z_{21} = 2 \Omega$$

$$\frac{V_2}{I_2} = 4$$

$$Z_{22} = 4 \Omega$$

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$$

put $V_2 = 0$ in eqn ②

$$0 = 4 I_2 + 2 I_1$$

$$-4 I_2 = 2 I_1$$

$$I_2 = -\frac{1}{2} I_1$$

sub I_2 in ①

$$V_1 = 4 I_1 + 2 \left(-\frac{1}{2} I_1 \right)$$

$$V_1 = 4 I_1 - I_1$$

$$V_1 = 3 I_1$$

$$\frac{V_1}{I_1} = 3 = \frac{I_1}{V_1} \cdot \frac{1}{3} V$$

$$Y_{11} = \frac{1}{3} S$$

$$I_1 = \frac{V_1}{3}$$

$$I_2 = -\frac{1}{2} \left(\frac{V_1}{3} \right)$$

$$\frac{I_2}{V_1} = -\frac{1}{6} S$$

put $V_1 = 0$ in eqn (1)

$$0 = 4 I_1 + 2 I_2$$

$$-4 I_1 = 2 I_2$$

$$I_2 = -2 I_1$$

sub I_2 in

$$V_2 = 2 I_1 + 4(-2 I_1)$$

$$V_2 = 2 I_1 - 8 I_1$$

$$V_2 = -6 I_1$$

$$\frac{V_2}{I_2} = -6 \quad \frac{I_2}{V_2} = -\frac{1}{6} = Y_{12}$$

$$Y_{12} = -\frac{1}{6}$$

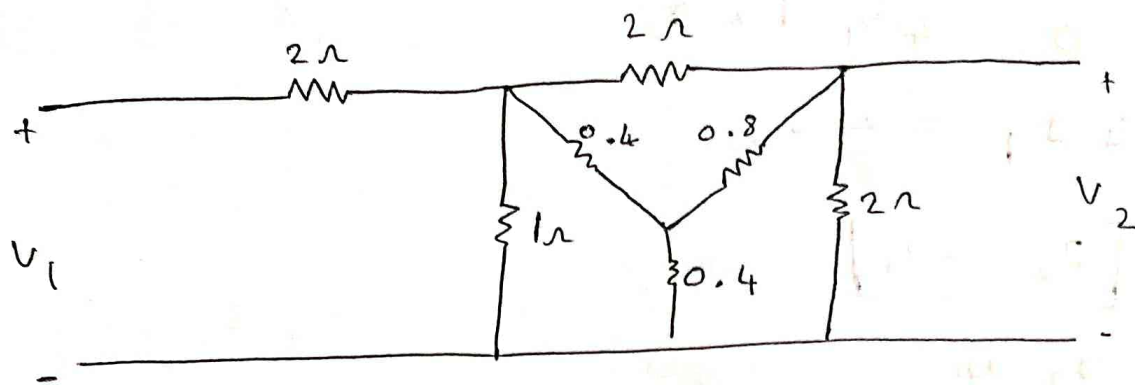
$$I_1 = \frac{V_2}{-6}$$

$$I_2 = \frac{2 V_2}{3}$$

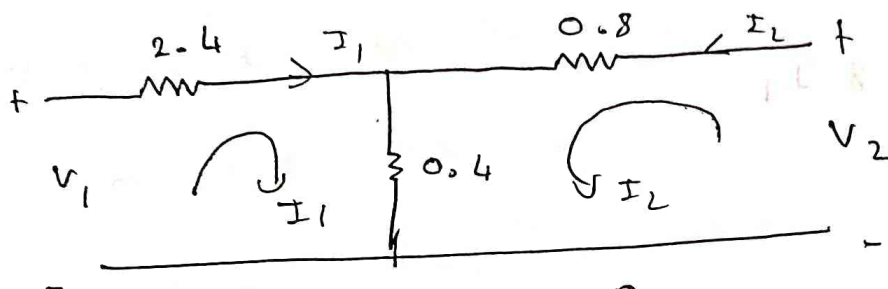
$$\frac{I_2}{V_2} = \frac{1}{3}$$

$$[Y] = \begin{bmatrix} \frac{1}{3} & -\frac{1}{6} \\ \frac{1}{6} & \frac{1}{3} \end{bmatrix}$$

2) Find Z_1 , Y_1 ABCD parameters for the ckt shown.



soln:



apply KVL to loop ①

$$-2.4 I_1 - 0.4 I_1 - 0.4 I_2 + V_1 = 0$$

$$V_1 = 2.8 I_1 + 0.4 I_2 \quad \text{--- ①}$$

apply KVL to loop ②

$$-0.8 I_2 - 0.4 I_2 - 0.4 I_1 + V_2 = 0$$

$$V_2 = 0.4 I_1 + 1.2 I_2 \quad \text{--- ②}$$

Z parameters

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

put $I_2 = 0$ in ① & ②

$$\frac{V_1}{I_1} = 2.8$$

$$Z_{11} = 2.8 \Omega$$

$$\frac{V_2}{I_1} = Z_{21} = 0.4 = Z_{21} = 0.4$$

put $I_1 = 0$

$$\frac{V_1}{I_2} = 0.4$$

$$\frac{V_2}{I_2} = 1.2$$

$$Z_{12} = 0.4$$

$$Z_{22} = 1.2$$

$$[Z] = \begin{bmatrix} 2.8 & 0.4 \\ 0.4 & 1.2 \end{bmatrix}$$

$$\Delta Z = 2.8(1.2) - (0.4)(0.4)$$

$$\Delta Z = 3.2$$

$$\begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} = \begin{bmatrix} \frac{Z_{22}}{\Delta Z} & -\frac{Z_{12}}{\Delta Z} \\ -\frac{Z_{21}}{\Delta Z} & \frac{Z_{11}}{\Delta Z} \end{bmatrix}$$

$$[y] = \begin{bmatrix} \frac{1.2}{3.2} & -\frac{0.4}{3.2} \\ -\frac{0.4}{3.2} & \frac{2.8}{3.2} \end{bmatrix}$$

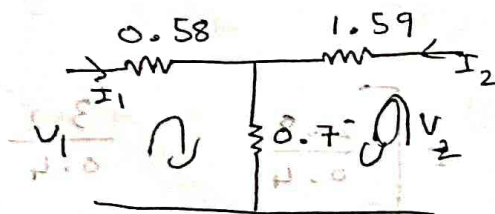
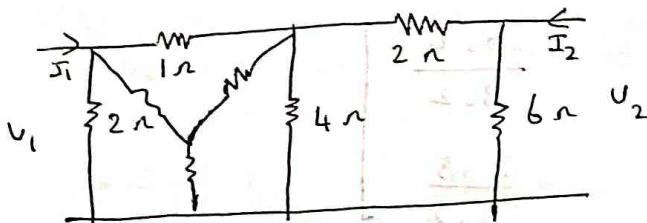
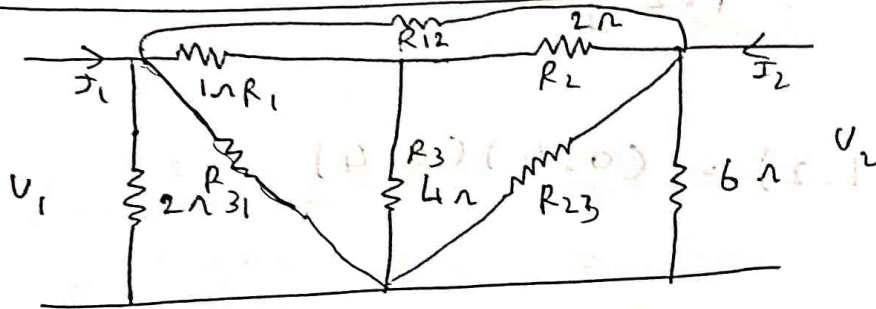
$$[y] = \begin{bmatrix} 0.375 & -0.125 \\ -0.125 & 0.875 \end{bmatrix}$$

$$[ABCD] = \begin{bmatrix} \frac{Z_{11}}{Z_{21}} & \frac{\Delta Z}{Z_{21}} \\ \frac{1}{Z_{21}} & \frac{Z_{22}}{Z_{21}} \end{bmatrix} = \begin{bmatrix} \frac{2.8}{0.4} & \frac{3.2}{0.4} \\ \frac{1}{0.4} & \frac{1.2}{0.4} \end{bmatrix}$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 7 & 8 \\ 2.5 & 3 \end{bmatrix}$$

$$\therefore A = 7, \quad B = 8, \quad C = 2.5, \quad D = 3$$

3)



Z parameters

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

apply KVL to loop ①

$$-0.58 I_1 - 0.7 I_1 - 0.7 I_2 + V_1 = 0$$

$$-1.28 I_1 - 0.7 I_2 + V_1 = 0$$

$$\boxed{V_1 = 1.28 I_1 + 0.7 I_2} \quad \text{--- ①}$$

apply KVL to loop ②

$$-1.59 I_2 - 0.7 I_2 - 0.7 I_1 + V_2 = 0$$

$$\boxed{V_2 = 0.7 I_1 + 2.29 I_2} \quad \text{--- ②}$$

put $I_2 = 0$ in ① & ②

$$\frac{V_1}{I_1} = 1.28 \Rightarrow \boxed{Z_{11} = 1.28}$$

$$\frac{V_2}{I_1} = 0.7 \Rightarrow \boxed{Z_{21} = 0.7}$$

put $I_1 = 0$ in ① & ②

$$\frac{V_1}{I_2} = 0.7 \Rightarrow \boxed{Z_{12} = 0.7}$$

$$\frac{V_2}{I_2} = 2.29 \Rightarrow \boxed{Z_{22} = 2.29}$$

$$\therefore [Z] = \begin{bmatrix} 1.28 & 0.7 \\ 0.7 & 2.29 \end{bmatrix}$$

Y parameters:

$$I_1 = Y_{11} V_1 + Y_{12} V_2$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2$$

put $V_2 = 0$ in eqn ②

$$0 = 0.7 I_1 + 2.29 I_2$$

$$I_2 = -\frac{0.7}{2.29} I_1$$

$$\boxed{I_2 = -0.307 I_1}$$

subs I_2 in ①

$$V_1 = 1.28 I_1 + 0.7 I_2$$

$$V_1 = 1.28 I_1 + 0.7(-0.3) I_1$$

$$\frac{V_1}{I_1} = \frac{1.07}{1.07} \Rightarrow \frac{I_1}{V_1} = 0.93$$

$$\boxed{Y_{11} = 0.93}$$

$$I_1 = \frac{V_1}{0.93}$$

$$I_2 = 0.305 \left(\frac{V_1}{0.93} \right)$$

$$\frac{I_2}{V_1} = \frac{0.305}{0.93} = \frac{-0.30}{1.070} = -0.284$$

$$\boxed{Y_{21} = -0.28 \text{ u}}$$

subs $V_1 = 0$ in eqn ① & ②

$$0 = 1.28 I_1 + 0.70 I_2$$

$$I_2 = -\frac{1.28}{0.70} I_1$$

$$I_2 = -1.82 I_1$$

subs I_2 in ②

$$V_2 = 0.70 I_1 + 2.28 (-1.82) I_1$$

$$V_2 = 0.70 I_1 - 4.14 I_1$$

$$V_2 = -3.44 I_1$$

$$\boxed{I_1 = -0.29 \text{ u} = Y_{12}}$$

$$V_2 = -3.44 I_1$$

$$I_2 = \frac{-1.52}{-3.44} V_2$$

$$\boxed{\frac{I_2}{V_2} = 0.52 \text{ u}}$$

$$h = \begin{bmatrix} 0.93 & -0.29 \\ -0.28 & 0.52 \end{bmatrix}$$

for h parameters:

$$V_1 = h_{11} I_1 + h_{12} V_2 \dots \textcircled{1}$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \dots \textcircled{2}$$

$$V_1 = 1.28 I_1 + 0.7 I_2 \dots \textcircled{3}$$

$$V_2 = 0.7 I_1 + 2.29 I_2 \dots \textcircled{4}$$

$$I_2 = \frac{V_2 - 0.7 I_1}{2.29}$$

$$I_2 = 0.43 V_2 - 0.3056 I_1 \dots \textcircled{5}$$

comparing ⑤ with ②

$$h_{21} = -0.30$$

$$h_{22} = 0.43$$

sub I_2 in V_1

$$V_1 = 1.28 I_1 + 0.7 \left(\frac{V_2 - 0.7 I_1}{2.29} \right)$$

$$V_1 = 1.28 I_1 + \frac{0.7 V_2}{2.29} - 0.30 I_1$$

$$V_1 = 0.30 V_2 + 0.98 I_1 \dots \textcircled{6}$$

comparing ⑥ & ①

$$h_{11} = 0.98$$

$$h_{12} = 0.30$$

$$[h] = \begin{bmatrix} 0.98 & 0.30 \\ -0.30 & 0.43 \end{bmatrix}$$

ABCD

$$V_1 = 1.28 I_1 + 0.70 I_2 \quad \text{①}$$

$$V_2 = 0.70 I_1 + 2.28 I_2 \quad \text{②}$$

$$V_1 = A V_2 - B I_2 \quad \text{③}$$

$$I_1 = C V_2 - D I_2 \quad \text{④}$$

from ②

$$I_1 = \frac{V_2 - 2.28 I_2}{0.7} \quad \text{⑤} \quad \text{comparing ⑤ \& ④}$$

$$\boxed{I_1 = C = 1.42}$$

$$\boxed{D = 3.25}$$

Subs I_1 in V_1

$$V_1 = 1.28 (1.42 V_2 - 3.25 I_2) + 0.70 I_2$$

$$\boxed{V_1 = 1.81 V_2 - 3.45 I_2}$$

$$\boxed{A = 1.83}$$

$$\boxed{B = 3.45}$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1.83 & 3.45 \\ 1.42 & 3.25 \end{bmatrix}$$

Note :

1) For a Reciprocal network

$$Z_{12} = Z_{21}$$

$$Y_{12} = Y_{21}$$

$$h_{12} = -h_{21}$$

2) For a symmetrical network

$$Z_{11} = Z_{22}$$

$$Y_{11} = Y_{22}$$

$$A = D$$

• Applications of h , Z , T are

h parameter: It is used in analysis of small signal amplifier using BJT

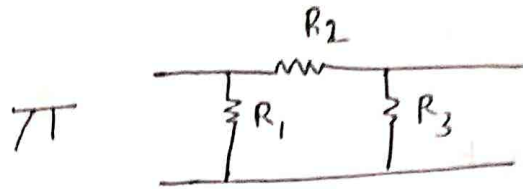
The equivalent circuit of h parameter can be used as an AC equivalent ckt of a small signal BJT amplifier to carry out AC analysis of amplifier

Z parameter: It is used to represent given two port network into equivalent T N/w which is the most important n/w in communication n/w

T parameter: They are generally used for analysis of power transmission line where i/p port is a sending end and o/p port of the n/w is the receiving end of the line

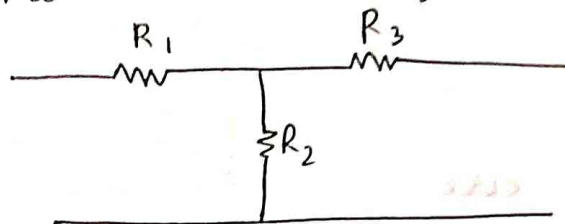
• If there is no dependent source in the given n/w then $z_{12} = z_{21}$ & $y_{12} = y_{21}$ if a n/w is said to be reciprocal

• If the n/w is of π type



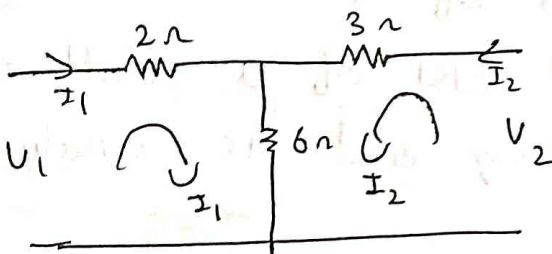
Then apply KCL

• If the n/w is of T Type



The apply KVL

→ find z and y parameter for n/w shown:



z parameter

$$V_1 = z_{11} I_1 + z_{12} I_2 \quad \dots \quad (1)$$

$$V_2 = z_{21} I_1 = z_{22} I_2 \quad \dots \quad (2)$$

apply KVL to loop (1)

$$-2 I_1 - 6(I_1 + I_2) + V_1 = 0$$

$$V_1 = 8 I_1 + 6 I_2 \quad \dots \quad (3)$$

apply KVL to loop (2)

$$-3 I_2 - 6 (I_2 + I_1) + V_2 = 0$$

$$V_2 = 6 I_1 + 9 I_2 \dots (4)$$

$$\text{put } I_2 = 0 \text{ in } (3) \text{ \& } (4)$$

$$\frac{V_1}{I_1} = 8 \Rightarrow \boxed{Z_{11} = 8 \Omega}$$

$$\frac{V_2}{I_2} = 6 \Rightarrow \boxed{Z_{22} = 6}$$

$$\text{put } I_1 = 0 \text{ in } (3) \text{ \& } (4)$$

$$\frac{V_2}{I_2} = 9 \Rightarrow \boxed{Z_{22} = 9}$$

$$\frac{V_1}{I_2} = 6 \Rightarrow \boxed{Z_{12} = 6}$$

$$[Z] = \begin{bmatrix} 8 & 6 \\ 6 & 9 \end{bmatrix}$$

$$\text{put } V_2 = 0 \text{ in } (4)$$

$$-6 I_1 + 9 I_2$$

$$I_2 = -\frac{6}{9} I_1$$

$$I_2 = -0.66 I_1$$

$$\text{subs } I_2 \text{ in } (3)$$

$$V_1 = 8 I_1 + 6 (-0.66) I_1$$

$$\frac{V_1}{I_1} = 4.04$$

$$\frac{I_1}{V_1} = \frac{1}{4.04} = 0.25 = Y_{11}$$

$$\boxed{\Delta Z = 36}$$

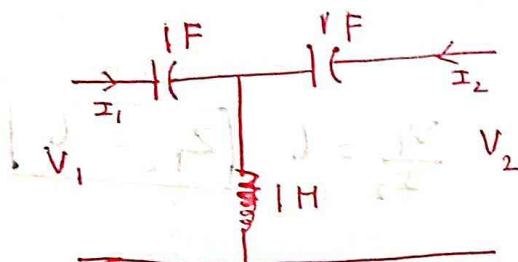
$$Y_{12} = -\frac{Z_{12}}{\Delta Z} = -\frac{6}{36} = -0.16 \Omega^{-1}$$

$$Y_{21} = -\frac{Z_{21}}{\Delta Z} = -\frac{6}{36} = -0.16 \Omega^{-1}$$

$$Y_{22} = \frac{Z_{11}}{\Delta Z} = \frac{8}{36} = 0.22$$

$$[Y] = \begin{bmatrix} 0.25 & -0.16 \\ -0.16 & 0.22 \end{bmatrix}$$

→ Find $Z, Y, h, ABCD$

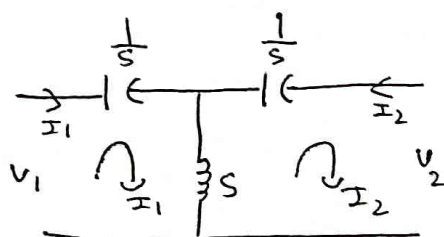


laplace transform of

$$IF \text{ is } \frac{1}{s}$$

$$IH \text{ is } s$$

soln:



Z parameters

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad \dots \textcircled{1}$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad \dots \textcircled{2}$$

apply KVL to loop ①

$$V_1 - \frac{1}{s} I_1 - s(I_1 + I_2) = 0$$

$$V_1 - \frac{1}{s} I_1 - s I_1 - s I_2 = 0$$

$$V_1 = \left(\frac{1 + s^2}{s} \right) I_1 + s I_2 \quad \dots \textcircled{3}$$

apply KVL to loop 2

$$-\frac{1}{s} I_2 - s I_2 - s I_1 + V_2 = 0$$

$$V_2 = s I_1 + \left(\frac{1 + s^2}{s} \right) I_2 \quad \dots \textcircled{4}$$

put $I_2 = 0$ in (3) & (4)

$$\frac{V_1}{I_1} = \boxed{Z_{11} = \frac{1+S^2}{S}} \quad \left| \quad \frac{V_2}{I_1} = S \quad \boxed{Z_{21} = S} \right.$$

put $I_1 = 0$ in (3) & (4)

$$\frac{V_1}{I_2} = \boxed{Z_{12} = S} \quad \left| \quad \frac{V_2}{I_2} = \frac{1+S^2}{S} \quad \boxed{Z_{22} = \frac{1+S^2}{S}} \right.$$

$$Z = \begin{bmatrix} \frac{1+S^2}{S} & S \\ S & \frac{1+S^2}{S} \end{bmatrix}$$

9 Parameters are given by

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \quad \dots \quad (1)$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \quad \dots \quad (2)$$

$$V_1 = \left(\frac{1+S^2}{S} \right) I_1 + S I_2 \quad \dots \quad (3)$$

$$V_2 = S I_1 + \left(\frac{1+S^2}{S} \right) I_2 \quad \dots \quad (4)$$

put $V_2 = 0$ in eqn (4)

$$0 = S I_1 + \left(\frac{1+S^2}{S} \right) I_2$$

$$S I_1 = - \frac{1+S^2}{S} I_2$$

$$I_1 = - \left(\frac{1+S^2}{S^2} \right) I_2$$

$$I_2 = \frac{-S^2}{1+S^2} I_1$$

$$\frac{2+2}{2+1} = \frac{4}{3} = \frac{4}{3}$$

$$V_1 = \left(\frac{1+s^2}{s} \right) \cdot - \left(\frac{1+s^2}{s^2} \right) I_2 + s I_2$$

$$V_1 = - \frac{(1+s^2)^2}{s^3} I_2 + s I_2$$

$$V_1 = \frac{s^4 - (1+s^2)^2}{s^3} I_2$$

$$V_1 = \frac{s^4 - (1+s^4 + 2s^2)}{s^3} I_2$$

$$V_1 = \frac{\cancel{s^4} - 1 - \cancel{s^4} - 2s^2}{s^3} I_2$$

$$\frac{I_2}{V_1} = - \frac{s^3}{(1+2s^2)}$$

$$y_{21} = \frac{-s^3}{1+2s^2}$$

sub I_2 in eqn (3)

$$V_1 = \left(\frac{1+s^2}{s} \right) I_1 + s \left(\frac{-s^2}{1+s^2} \right) I_1$$

$$V_1 = \left(\frac{1+s^2}{s} - \frac{s^3}{1+s^2} \right) I_1$$

$$V_1 = \frac{(1+s^2)^2 - s^4}{s(1+s^2)} I_1$$

$$V_1 = \frac{1 + \cancel{s^4} + 2s^2 - \cancel{s^4}}{s + s^3} I_1$$

$$y_{11} = \frac{I_1}{V_1} = \frac{s + s^3}{1+2s^2}$$

put $V_1 = 0$

$$-\frac{1+s^2}{s} I_1 = s I_2$$

$$I_1 = -\frac{1+s^2}{s^2} I_2$$

$$I_1 = -\frac{s^2}{1+s^2} I_2$$

subs I_1 in eqn (4)

$$V_2 = s \left(-\frac{s^2}{1+s^2} \right) I_2 + \left(\frac{1+s^2}{s^2} \right) I_2$$

$$V_2 = -\frac{s^3}{1+s^2} I_2 + \frac{1+s^2}{s^2} I_2$$

$$V_2 = -\frac{s^4 + (1+s^2)^2}{s + s^3} I_2$$

$$V_2 = -\frac{s^4 + 1 + 2s^2}{s + s^3} I_2$$

$$V_2 = \frac{1 + 2s^2}{s + s^3} I_2$$

$$Y_{22} = \frac{I_2}{V_2} = \frac{s + s^3}{1 + 2s^2}$$

subs I_2 in eqn (4)

$$V_2 = \frac{1+s^2}{s} I_1 + s \left(-\frac{1+s^2}{s^2} \right) I_1$$

$$V_2 = \left(\frac{1+s^2}{s} - \frac{s^3}{1+s^2} \right) I_1$$

$$V_2 = \frac{(1+s^2)I_1 - sI_1}{s^3}$$

$$V_2 = -\frac{s^4 - 1 - s^4 - 2s^2}{s^3} I_1$$

$$\boxed{\frac{I_1}{V_2} = \frac{-s^3}{1+2s^2}}$$

$$Y = \begin{bmatrix} \frac{s+s^3}{1+2s^2} & \frac{-s^3}{1+2s^2} \\ \frac{-s^3}{1+2s^2} & \frac{s+s^3}{1+2s^2} \end{bmatrix}$$

→ h parameters

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \text{--- (1)}$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \text{--- (2)}$$

$$V_1 = \frac{1+s^2}{s} I_1 + s I_2 \quad \text{--- (3)}$$

$$V_2 = s I_1 + \frac{1+s^2}{s} I_2 \quad \text{--- (4)}$$

from (4)

$$I_2 = \frac{V_2 - s I_1}{\frac{1+s^2}{s}}$$

$$= \frac{s V_2}{1+s^2} - \frac{s^2 I_1}{1+s^2}$$

$$I_2 = \frac{s}{1+s^2} V_2 - \frac{s^2}{1+s^2} I_1 \quad \text{--- (5)}$$

comparing (5) & (2) we get.

$$\boxed{h_{21} = \frac{-s^2}{1+s^2}}$$

$$\boxed{h_{22} = \frac{s}{1+s^2}}$$

sub $\pm_2$ in (3)

$$V_1 = \frac{1+s^2}{s} I_1 + s \left(\frac{s}{1+s^2} V_2 - \frac{s^2}{1+s^2} I_1 \right)$$

$$V_1 = \frac{1+s^2}{s} I_1 + \left(\frac{s^2}{1+s^2} V_2 - \frac{s^3}{1+s^2} I_1 \right)$$

$$V_1 = \frac{s^2}{1+s^2} V_2 + \left(\frac{1+s^2}{s} - \frac{s^3}{1+s^2} \right) I_1$$

$$V_1 = \frac{s^2}{1+s^2} V_2 + \left(\frac{1+s^2 - s^4}{s+s^3} \right) I_1$$

$$V_1 = \frac{s^2}{1+s^2} V_2 + \frac{1+2s^2}{s+s^3} I_1 \quad \dots (6)$$

comparing (6) in (5)

$$h_{11} = \frac{1+2s^2}{s+s^3}$$

$$h_{12} = \frac{s^2}{1+s^2}$$

$$[h] = \begin{bmatrix} \frac{1+2s^2}{s+s^3} & \frac{s^2}{1+s^2} \\ \frac{-s^2}{1+s^2} & \frac{s}{1+s^2} \end{bmatrix}$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \frac{z_{11}}{z_{21}} & \frac{\Delta z}{z_{21}} \\ \frac{1}{z_{21}} & \frac{z_{22}}{z_{21}} \end{bmatrix}$$

$$\Delta z = \frac{1+s^2}{s} \left(\frac{1+s^2}{s} \right) - s(s)$$

$$\Delta z = \left(\frac{1+s^2}{s} \right)^2 - s^2$$

$$\Delta z = \frac{1+2s^2}{s^2} - s^2$$

$$\Delta z = \frac{1+2s^2}{s^2}$$

$$A = \frac{Z_{11}}{Z_{21}} = \frac{1+s^2}{s^2}$$

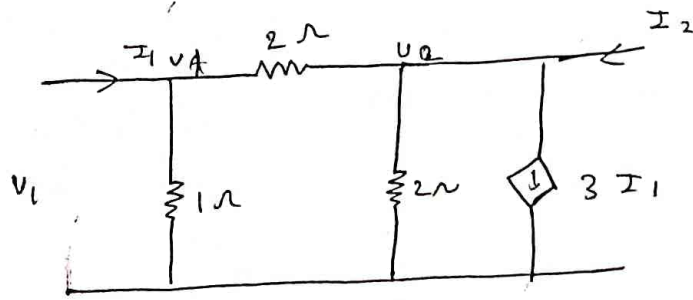
$$B = \frac{\Delta Z}{Z_{21}} = \frac{1+s^2}{s^3}$$

$$C = \frac{1}{Z_{21}} = \frac{1}{s}$$

$$D = \frac{Z_{22}}{Z_{21}} = \frac{1+s^2}{s^2}$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \frac{1+s^2}{s^2} & \frac{1+s^2}{s^3} \\ \frac{1}{s} & \frac{1+s^2}{s^2} \end{bmatrix}$$

* Find z and y parameters.



y parameter

$$I_1 = y_{11} V_1 + y_{12} V_2 \quad \text{--- (1)}$$

$$I_2 = y_{21} V_1 + y_{22} V_2 \quad \text{--- (2)}$$

apply KCL at node ①

$$V_A - V_2 + V_A + \frac{V_A - V_2}{2} = I_1$$

$$I_1 = V_1 + 0.5V_1 + 0.5V_2$$

$$I_1 = 1.5V_1 - 0.5V_2 \quad \text{--- (3)}$$

apply KCL at node ②

$$I_2 = \frac{V_2 - V_1}{2} + \frac{V_2}{2}$$

$$I_2 = 0.5V_2 - 0.5V_1 + 0.5V_2 + 3(1.5V_1 - 0.5V_2)$$

$$I_2 = -0.5V_1 + 4V_2 \quad \text{--- (4)}$$

comparing (3) & (1)

$$y_{11} = 1.5 \quad y_{12} = -0.5$$

comparing (4) & (2)

$$y_{21} = -0.5 \quad y_{22} = 4$$

$$y = \begin{bmatrix} 1.5 & -0.5 \\ 4 & -0.5 \end{bmatrix}$$

$$\Delta y = 1.5(-0.5) - (-0.5)(4)$$

$$\boxed{\Delta y = 1.25}$$

$$z_{11} = \frac{y_{22}}{\Delta y} = \frac{-0.5}{1.25} = -0.4 \Omega$$

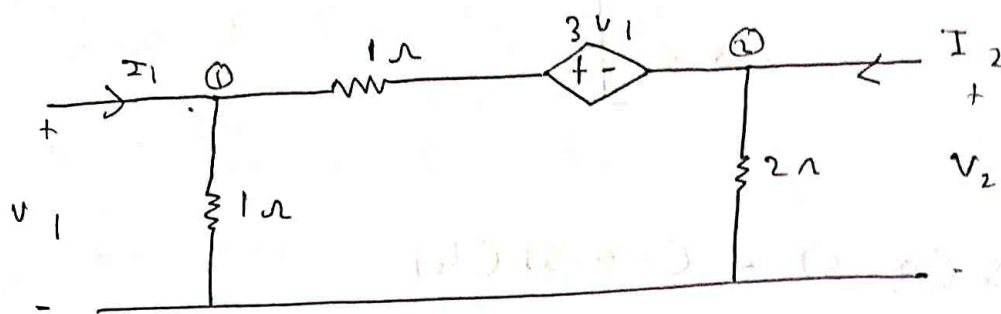
$$z_{12} = -\frac{y_{12}}{\Delta y} = -\frac{(-0.5)}{1.25} = 0.4 \Omega$$

$$z_{22} = \frac{y_{11}}{\Delta y} = \frac{1.5}{1.25} = 1.2 \Omega$$

$$z_{21} = -\frac{y_{21}}{\Delta y} = -\frac{4}{1.25} = -3.2 \Omega$$

$$z = \begin{bmatrix} -0.4 & 0.4 \\ -3.2 & 1.2 \end{bmatrix}$$

* Find y and z parameters for the ckt shown.



apply KCL at ①

$$I_1 = \frac{V_1}{1} + \frac{V_1 - 3V_1 - V_2}{1}$$

$$I_1 = V_1 + V_1 - 3V_1 - V_2$$

$$\boxed{I_1 = -1V_1 - 1V_2} \dots \textcircled{1}$$

apply KCL at ②

$$I_2 = \frac{V_2}{2} + \frac{V_2 + 3V_1 - V_1}{1}$$

$$I_2 = (0.5 + 1)V_2 + 2V_1$$

$$\boxed{I_2 = 1.5V_2 + 2V_1} \dots \textcircled{2}$$

y Parameters

$$I_1 = Y_{11}V_1 + Y_{12}V_2 \dots \textcircled{3}$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2 \dots \textcircled{4}$$

comparing ③ & ①

$$Y_{11} = -1$$

$$Y_{12} = -1$$

comparing ④ & ②

$$Y_{21} = 2$$

$$Y_{22} = 1.5$$

$$\therefore [Y] = \begin{bmatrix} -1 & -1 \\ 2 & 1.5 \end{bmatrix}$$

$$[Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}$$

$$Z_{11} = \frac{Y_{22}}{\Delta Y} = \frac{1.5}{-0.5} = -3 \Omega$$

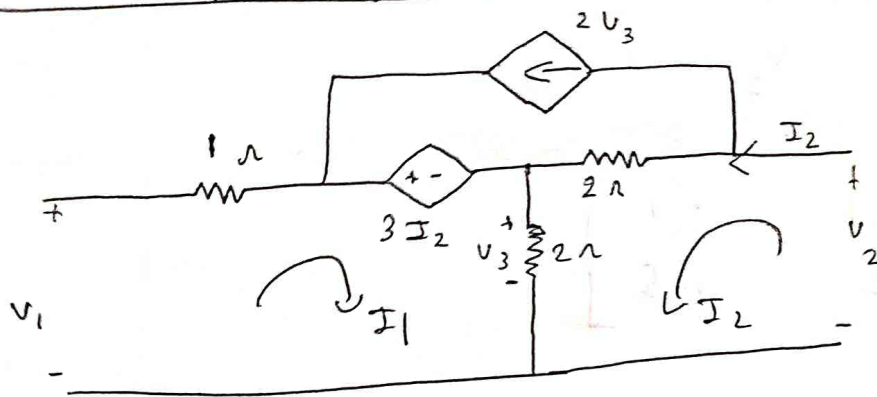
$$Z_{12} = \frac{-Y_{12}}{\Delta Y} = \frac{-(-1)}{-0.5} = -2 \Omega$$

$$Z_{21} = \frac{-Y_{21}}{\Delta Y} = \frac{2}{0.5} = 4 \Omega$$

$$Z_{22} = \frac{Y_{11}}{\Delta Y} = \frac{1}{0.5} = 2 \Omega$$

$$[Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} -3 & -2 \\ 4 & 2 \end{bmatrix}$$

* Find y & z for the circuit shown:



from ckt $I_3 = 2V_3$

$$V_3 = (I_1 + I_2) \cdot 2$$

$$V_3 = 2I_1 + 2I_2$$

apply KVL to loop ①

$$V_1 - 1I_1 - 3I_2 + (2I_1 + 2I_2) = 0$$

$$V_1 = 3I_1 + 5I_2 \quad \dots \quad \text{①}$$

apply KVL to loop ②

$$V_2 - 2(I_2 - I_3) - 2(I_1 + I_2) = 0$$

$$V_2 - 2I_2 + 2I_3 - 2I_1 - 2I_2 = 0$$

$$V_2 = -2I_2 + 2(2I_1 + 2I_2) - 2I_1 - 2I_2 = 0$$

$$V_2 = -2I_2 + 8I_1 + 8I_2 - 2I_1 - 2I_2 = 0$$

$$V_2 = -6I_1 - 4I_2 \quad \dots \quad \text{②}$$

2 ports.

$$V_1 = Z_{11}I_1 + Z_{12}I_2 \quad \dots \quad \text{③}$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2 \quad \dots \quad \text{④}$$

$$z_{11} = 3 \Omega$$

$$z_{12} = 5 \Omega$$

$$z_{21} = -6 \Omega$$

$$z_{22} = -4 \Omega$$

$$Z = \begin{bmatrix} 3 & 5 \\ -6 & -4 \end{bmatrix} \Omega$$

y parameter

$$Y = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} = \begin{bmatrix} \frac{z_{22}}{\Delta Z} & -\frac{z_{12}}{\Delta Z} \\ -\frac{z_{21}}{\Delta Z} & \frac{z_{11}}{\Delta Z} \end{bmatrix}$$

$$\Delta Z = 3(-4) - (5(-6))$$

$$\Delta Z = -12 + 30$$

$$\Delta Z = 18$$

$$y_{11} = \frac{z_{22}}{\Delta Z} = \frac{-4}{18} = -0.22 \Omega^{-1}$$

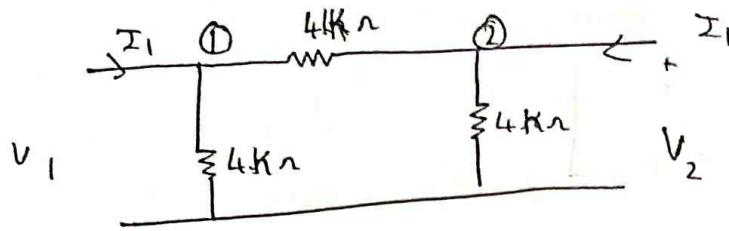
$$y_{12} = -\frac{z_{12}}{\Delta Z} = -\frac{5}{18} = -0.27 \Omega^{-1}$$

$$y_{21} = -\frac{z_{21}}{\Delta Z} = -\frac{-6}{18} = 0.33 \Omega^{-1}$$

$$y_{22} = \frac{z_{11}}{\Delta Z} = \frac{3}{18} = 0.166 \Omega^{-1}$$

$$Y = \begin{bmatrix} -0.22 & -0.27 \\ 0.33 & 0.166 \end{bmatrix} \Omega^{-1}$$

* Find y parameters for the ckt shown and use them to find transmission parameter.



$$I_1 = Y_{11} V_1 + Y_{12} V_2 \quad \dots \quad (1)$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \quad \dots \quad (2)$$

$$\begin{bmatrix} 0.5 \text{ m} & -0.25 \text{ m} \\ -0.25 \text{ m} & 0.5 \text{ m} \end{bmatrix}$$

apply KCL at node ①

$$I_1 = \frac{1}{4\text{k}} V_1 + \frac{V_1 - V_2}{4\text{k}}$$

$$I_1 = \left(\frac{1}{4\text{k}} + \frac{1}{4\text{k}} \right) V_1 - 0.25 \text{ m} V_2$$

$$I_1 = \frac{2}{4\text{k}} V_1 - 0.25 \text{ m} V_2$$

$$I_1 = 0.5 \text{ m} V_1 - 0.25 \text{ m} V_2 \quad \dots \quad (3)$$

apply KCL at node ②

$$I_2 = \frac{V_2}{4\text{k}} + \frac{V_2 - V_1}{4\text{k}}$$

$$I_2 = -\frac{V_1}{4\text{k}} + \frac{2}{4\text{k}} V_2 \quad \dots$$

$$I_2 = -0.25 \text{ m} V_1 + 0.5 \text{ m} V_2 \quad \dots \quad (4)$$

comparing ① & ③, ② & ④

$$y = \begin{bmatrix} 0.5 \text{ m} & -0.25 \text{ m} \\ -0.25 \text{ m} & 0.5 \text{ m} \end{bmatrix}$$

$$A = -\frac{y_{22}}{y_{21}} = -\frac{0.5 \text{ m}}{-0.25 \text{ m}} = +2$$

$$B = -\frac{1}{y_{21}} = -\frac{1}{-0.25 \text{ m}} = 4 \text{ K}$$

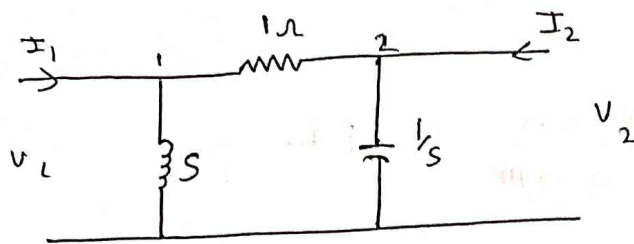
$$C = -\frac{\Delta y}{y_{21}} = -\frac{0.1875 \text{ m}}{-0.125 \text{ m}} = 0.75$$

$$D = -\frac{y_{11}}{y_{21}} = -\frac{0.5 \text{ m}}{-0.25 \text{ m}} = +2$$

$$[ABCD] = \begin{bmatrix} +2 & 4 \text{ K} \\ 0.75 & +2 \end{bmatrix}$$

* Determine transmission parameters for the

n/w shown.



apply Kc at ①

$$\frac{V_1}{s} + \frac{V_1 - V_2}{1} = I_1$$

$$I_1 = \frac{1}{s} V_1 + 1 V_1 - 1 V_2$$

$$I_1 = 1 + \frac{1}{s} V_1 - 1 V_2$$

$$I_1 = \frac{s+1}{s} V_1 - 1 V_2 \quad \dots \quad \textcircled{1}$$

Apply Kc at node ②

$$I_2 = \frac{V_2}{1/s} + V_2 - V_1$$

$$I_2 = s V_2 + V_2 - V_1$$

$$V_1 = (1+s) V_2 - 1 I_2 \quad \dots \quad \textcircled{2}$$

subs ② in ①

$$I_1 = \frac{s+1}{s} [(1+s) V_2 - 1 I_2] - 1 V_2 = \frac{(s+1)^2}{s} V_2 - \left(\frac{s+1}{s}\right) I_2$$

$$I_1 = \frac{s^2 + 1 + 2s}{s} V_2 - \left(\frac{s+1}{s}\right) I_2 = \boxed{\frac{s^2 + 1 + 2s}{s} V_2 - \frac{s+1}{s} I_2}$$

$$I_1 = \frac{s^2 + 1 + 2s}{s} V_2 - \frac{s+1}{s} I_2 \quad \dots \quad \textcircled{3}$$

ABCD parameters:

$$V_1 = A V_2 - B I_2 \quad \dots \quad \textcircled{4}$$

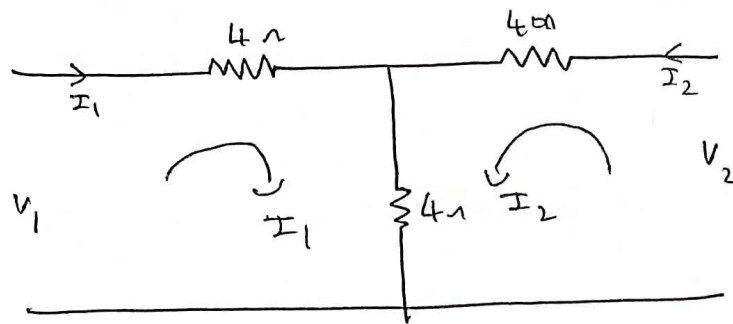
$$I_1 = C V_2 - D I_2 \quad \dots \quad \textcircled{5}$$

comparing ④ & ②

comparing ⑤ & ③

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} s+1 & 1 \\ \frac{s^2 + 2s + 1}{s} & \frac{s+1}{s} \end{bmatrix}$$

* Determine H parameter for the ckt.



apply KVL at loop ①

$$V_1 - 4 I_1 - 4 I_1 - 4 I_2 = 0$$

$$\boxed{V_1 = 8 I_1 + 4 I_2} \dots \textcircled{1}$$

apply KVL to loop ②

$$V_2 - 4 I_2 - 4 I_2 - 4 I_1 = 0$$

$$\boxed{V_2 = 8 I_2 + 4 I_1} \dots \textcircled{2}$$

$$I_2 = \frac{V_2 - 4 I_1}{8}$$

$$\boxed{I_2 = 0.125 V_2 - 0.5 I_1} \dots \textcircled{3}$$

subst I_2 in ①

$$V_1 = 8 I_1 + 4 (0.125 V_2 - 0.5 I_1)$$

$$V_1 = 8 I_1 + 0.5 V_2 - 2 I_1$$

$$\boxed{V_1 = 6 I_1 + 0.5 V_2} \dots \textcircled{4}$$

comparing ④ & ⑤

comparing ③ & ⑥

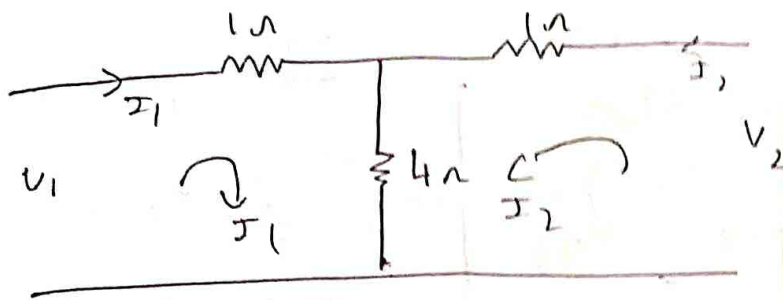
$$[h] = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} 6 & 0.5 \\ -0.5 & 0.125 \end{bmatrix}$$

h parameters

$$V_1 = h_{11} I_1 + h_{12} V_2 \dots \textcircled{5}$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \dots \textcircled{6}$$

Determine ABCD parameters for the circuit.



ABCD parameters

$$V_1 = AV_2 - BI_2 \quad \text{--- (1)}$$

$$I_1 = CV_2 - DI_2 \quad \text{--- (2)}$$

apply KVL to loop ①,

$$V_1 - I_1 - 4(I_1 + I_2) = 0$$

$$V_1 - 5I_1 - 4I_2 = 0$$

$$\boxed{V_1 = 5I_1 + 4I_2} \quad \text{--- (3)}$$

$$V_2 - I_2 - 4(I_1 + I_2) = 0$$

$$V_2 - 5I_2 = 4I_1$$

$$\boxed{I_1 = 0.25V_2 - 1.25I_2} \quad \text{--- (4)}$$

subs (4) in (3)

$$V_1 = 5(0.25V_2 - 1.25I_2) + 4I_2$$

$$V_1 = 1.25V_2 - 6.25I_2 + 4I_2$$

$$\boxed{V_1 = 1.25V_2 - 2.25I_2} \quad \text{--- (5)}$$

comparing (5) & (1)

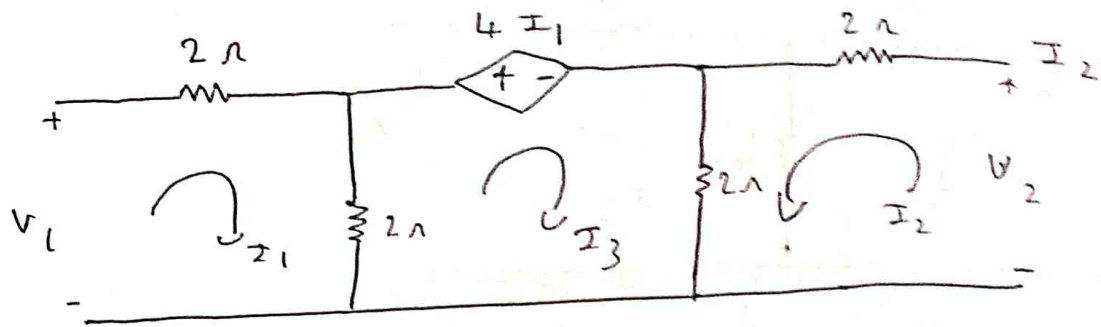
$$A = 1.25 \quad B = 2.25$$

comparing (4) & (2)

$$C = 0.25 \quad D = -1.25$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1.25 & 2.25 \\ 0.25 & -1.25 \end{bmatrix}$$

→ find 2 parameters for the circuit shown



apply KVL to loop ①

$$V_1 - 2I_1 - 2I_1 + 2I_3 = 0$$

$$\boxed{V_1 = 4I_1 - 2I_3}$$

apply KVL to loop ②

$$-2I_2 - 2(I_2 + I_3) + V_2 = 0$$

$$-2I_2 - 2I_2 - 2I_3 + V_2 = 0$$

$$\boxed{V_2 = +4I_2 + 2I_3}$$

apply KVL to loop ③

$$-4I_1 - 2(I_2 + I_3) - 2(I_3 - I_1) = 0$$

$$-4I_1 - 2I_2 - 2I_3 - 2I_3 + 2I_1 = 0$$

$$\underline{-2I_1 - 2I_2 - 4I_3 = 0}$$

2

$$-I_1 - I_2 - 2I_3 = 0$$

$$\boxed{2I_3 = -I_1 - I_2}$$

subs $2 I_3$ in V_1

$$V_1 = 4 I_1 - (-I_1 - I_2)$$

$$\boxed{V_1 = 5 I_1 + 1 I_2}$$

subs $2 I_3$ in V_2

$$V_2 = +4 I_2 + (-I_1 - I_2)$$

$$\boxed{V_2 = -1 I_1 + 3 I_2}$$

$$Z = \begin{bmatrix} 5 & 1 \\ -1 & 3 \end{bmatrix}$$

h parameters:

$$[h] = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} \frac{\Delta Z}{Z_{22}} & \frac{Z_{12}}{Z_{22}} \\ -\frac{Z_{21}}{Z_{22}} & \frac{1}{Z_{22}} \end{bmatrix}$$

$$h_{11} = \frac{\Delta Z}{Z_{22}} = \frac{16}{3} = 5.33$$

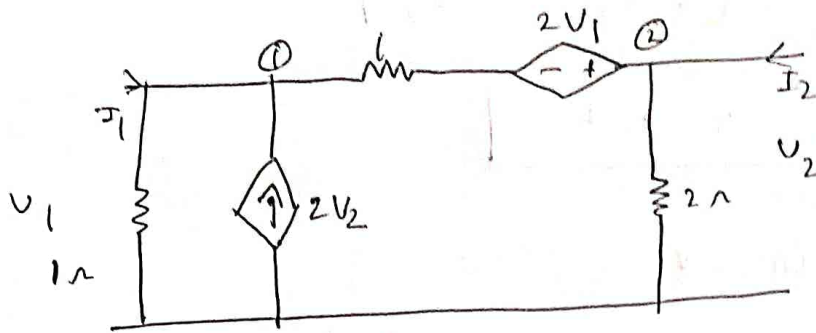
$$h_{12} = \frac{Z_{12}}{Z_{22}} = \frac{1}{3} = 0.33$$

$$h_{21} = -\frac{Z_{21}}{Z_{22}} = \frac{+1}{3} = 0.33$$

$$h_{22} = \frac{1}{Z_{22}} = \frac{1}{3} = 0.33$$

$$\therefore [h] = \begin{bmatrix} 5.33 & 0.33 \\ 0.33 & 0.33 \end{bmatrix}$$

→ Find y and z parameters for the n/w shown.



apply KCL at ①

$$V_1 + \underbrace{V_1 + 2V_1 - V_2}_1 = I_1 + 2V_2$$

$$\boxed{4V_1 - 3V_2 = I_1}$$

apply KCL at ②

$$\frac{V_2}{2} + V_2 - 2V_1 - V_1 = I_2$$

$$\boxed{1.5V_2 - 3V_1 = I_2}$$

~~put $V_2 = 0$~~

$$y_{11} = \frac{I_1}{V_1} = 4$$

$$x_{12} = \frac{V_1}{V_2} = -3$$

$$x_{21} = -3$$

$$y_{22} = 1.5$$

$$Y = \begin{bmatrix} 4 & -3 \\ -3 & 1.5 \end{bmatrix} \Omega$$

$$Z = \begin{bmatrix} \frac{1.5}{-3} & -\left(\frac{-3}{-3}\right) \\ -\left(\frac{-3}{-3}\right) & \frac{+4}{-3} \end{bmatrix}$$

$$Z = \begin{bmatrix} -0.5 & -1 \\ -1 & 1.33 \end{bmatrix} \Omega$$

→ prove that $AD - BC = 1$ for a reciprocal n/w

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \frac{Z_{11}}{Z_{21}} & \frac{\Delta Z}{Z_{21}} \\ \frac{1}{Z_{21}} & \frac{Z_{22}}{Z_{21}} \end{bmatrix}$$

$$A = \frac{Z_{11}}{Z_{21}}$$

$$B = \frac{\Delta Z}{Z_{21}}$$

$$C = \frac{1}{Z_{21}}$$

$$D = \frac{Z_{22}}{Z_{21}}$$

$$\Delta Z = Z_{11} Z_{22} - Z_{12} Z_{21}$$

$$= \begin{bmatrix} \frac{Z_{11}}{Z_{21}} & \frac{Z_{11} Z_{22} - Z_{12} Z_{21}}{Z_{21}} \\ \frac{1}{Z_{21}} & \frac{Z_{22}}{Z_{21}} \end{bmatrix}$$

$$= \frac{Z_{11} Z_{22}}{Z_{21}^2} - \frac{Z_{11} Z_{22} - Z_{21} Z_{12}}{Z_{21}^2}$$

$$= \frac{\cancel{Z_{11} Z_{22}} - \cancel{Z_{11} Z_{22}} + Z_{21} Z_{12}}{Z_{21}^2}$$

for a reciprocal n/w $Z_{21} = Z_{12}$

$$\therefore Z_{12} = Z_{21}$$

$$= \frac{\cancel{Z_{21}^2}}{\cancel{Z_{21}^2}}$$

$$\therefore AD - BC = 1$$

(18) For a symmetrical n/w show that $\Delta h = 1$

Soln:

$$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} \frac{\Delta Z}{Z_{22}} & \frac{Z_{12}}{Z_{22}} \\ \frac{-Z_{21}}{Z_{22}} & \frac{1}{Z_{22}} \end{bmatrix}$$

$$\Delta h = \frac{\Delta Z}{Z_{22}^2} - \frac{Z_{12}(-Z_{21})}{Z_{22}^2}$$

$$\Delta h = \frac{\Delta Z}{Z_{22}^2} + \frac{Z_{12} Z_{21}}{Z_{22}^2}$$

$$\Delta h = \frac{\Delta Z + Z_{12} Z_{21}}{Z_{22}^2}$$

$$\Delta Z = Z_{11} Z_{22} - Z_{12} Z_{21}$$

$$\Delta h = \frac{Z_{12} Z_{21} + Z_{11} Z_{22} - Z_{12} Z_{21}}{Z_{22}^2}$$

for a symmetrical n/w $Z_{11} = Z_{22}$

$$\therefore Z_{22} = Z_{11}$$

$$\Delta h = \frac{Z_{22} \cdot Z_{22}}{Z_{22}^2}$$

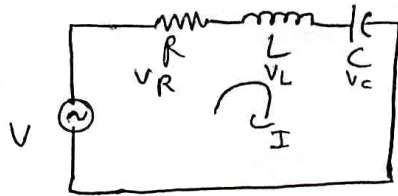
$$\Delta h = \frac{Z_{22}^2}{Z_{22}^2}$$

$$\boxed{\Delta h = 1}$$

11 hp

Resonance:

AC circuits which are made up of resistor, inductor, capacitor are said to be resonance circuits, when current drawn from the supply is in phase with the input voltage



→ Resonance:

It is a phenomenon or study of AC ckt's consisting of R, L & C of a variable frequency.

→ resonance are of two types

series & parallel resonance

current I is given by

$$I = \frac{V}{Z}$$

where Z is the impedance of ckt.

$$Z = R + j(\omega L - \omega C)$$

$$X_L = 2\pi f L$$

$$X_C = \frac{1}{2\pi f C}$$

Resonance is a ~~point~~ ^{condition} where $X_L = X_C$ for variable frequency.

at resonance $Z = R + j(\omega L - \omega C)$

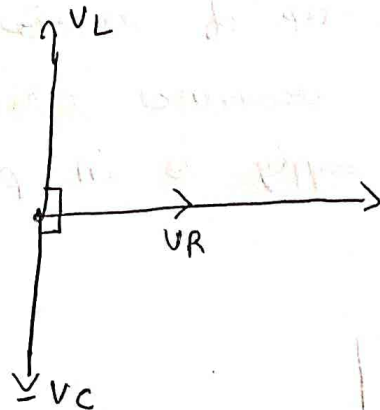
$$I_M = I_{max}$$

$$Z = R$$

Purely resistive
ckt

$$I_M = \frac{V}{R}$$

Phasor diagram



Exp for resonant frequency:

at Resonance

$$X_L = X_C$$

$$2\pi f L = \frac{1}{2\pi f C}$$

$$\Rightarrow \sqrt{f^2} = \frac{1}{\sqrt{4\pi^2 LC}} \quad f = \frac{1}{2\pi\sqrt{LC}}$$

$$f = \frac{1}{2\pi\sqrt{LC}}$$

- Resonance can be achieved by varying f by keeping L & C constant
- by varying L & C and keeping resonant frequency constant

→ Features of Resonance circuits:

- Inductive reactance = X_C (capacitive reactance)

$$X_L = X_C$$

- output current is in phase with the input voltage
- capacitive voltage = - of inductive voltage
- circuit works as purely resistive circuit
- power factor is unity

$$P = VI \cos \phi \quad \text{or} \quad \cos \phi = \frac{P}{VI} = 1$$

• Maximum current & minimum impedance.

1) A coil of 5 mH inductance & $10\ \Omega$ resistance is connected in series with $5\ \mu\text{F}$ capacitor, determine the frequency at which resonance.

given - $L = 5\text{ m}$

$C = 5\ \mu$

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$f_r = \frac{1}{2\pi\sqrt{5 \times 10^{-3} \times 5 \times 10^{-6}}}$$

$$\boxed{f_r = 1\text{ KHz}}$$

2) In a series RL C ckt driven with a sinusoidal AC voltage determine the value of capacitor required to achieve resonance at 5 KHz . if $R = 2\ \Omega$
 $L = 1\text{ mH}$

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

Square on both side

$$f_r^2 = \frac{1}{4\pi^2 LC}$$

$$C = \frac{1}{4\pi^2 L f_r^2}$$

$$C = \frac{1}{4 \times \pi^2 \times 1 \times 10^{-3} \times (5 \times 10^3)^2}$$

$$\boxed{C = 1.01\ \mu\text{F}}$$

→ Reactance curve:

The graph of individual reactance vs the frequency

$$X_L = \omega L$$

$$X_L \propto \omega$$

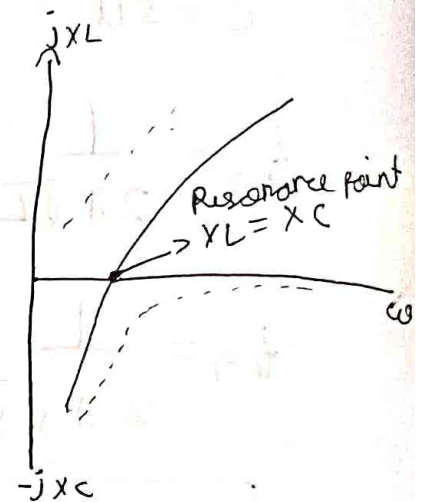
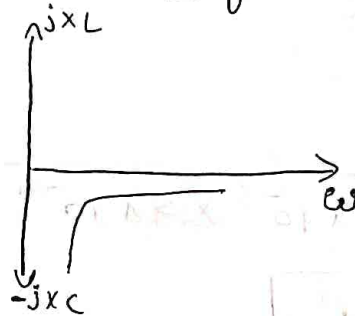
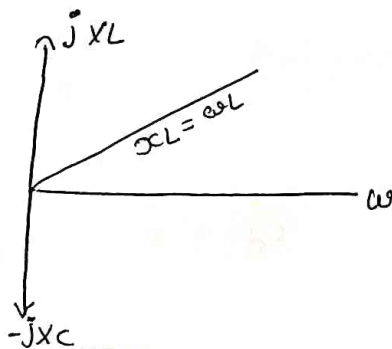
$$X_L \propto 2\pi f$$

$$X_C = \frac{1}{\omega C}$$

$$X_C \propto \frac{1}{\omega}$$

$$X_C = \frac{1}{2\pi f C}$$

$$X_L = X_C$$



• Variation of Impedance, admittance and current with frequency

$$\text{Let } Z = R + jX_L - jX_C$$

$$X_L$$

$$X_C$$

$$Z = R + j(X_L - X_C)$$

$$|Z| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

$$\text{at resonance } X_L = X_C$$

$$\therefore |Z| = \sqrt{R^2}$$

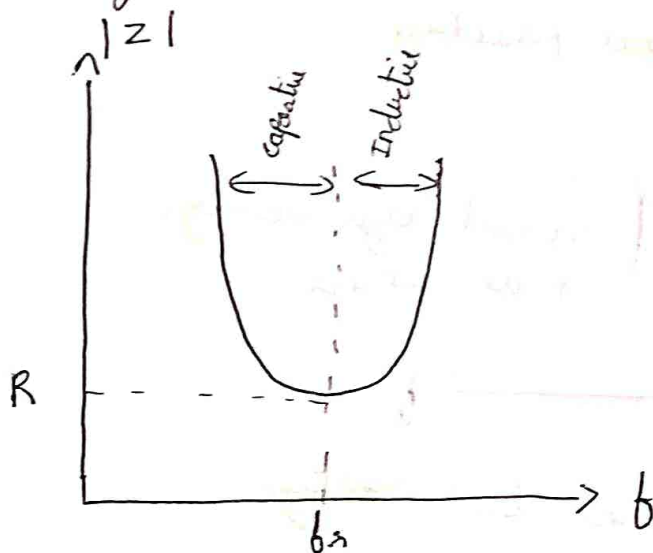
$$\boxed{|Z| = R}$$

At Resonance magnitude of impedance is equal to

R

$$|Z| = R$$

At frequencies other than f_r , the reactances are not equal to zero and magnitude of impedance is also greater than R

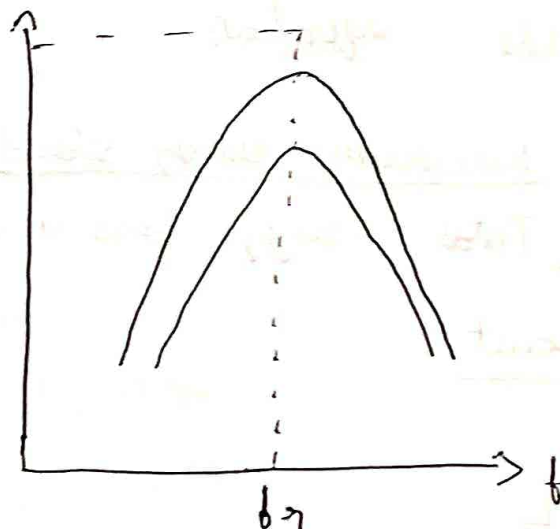


at series resonance impedance is minimum

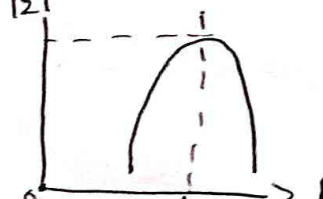
$$V = IR$$

as impedance is minimum the current flowing in the circuit

$$I = \frac{V}{Z}$$



since admittance is reciprocal of Z graph is shown below



• at frequencies below resonant frequency. ($f < f_r$)

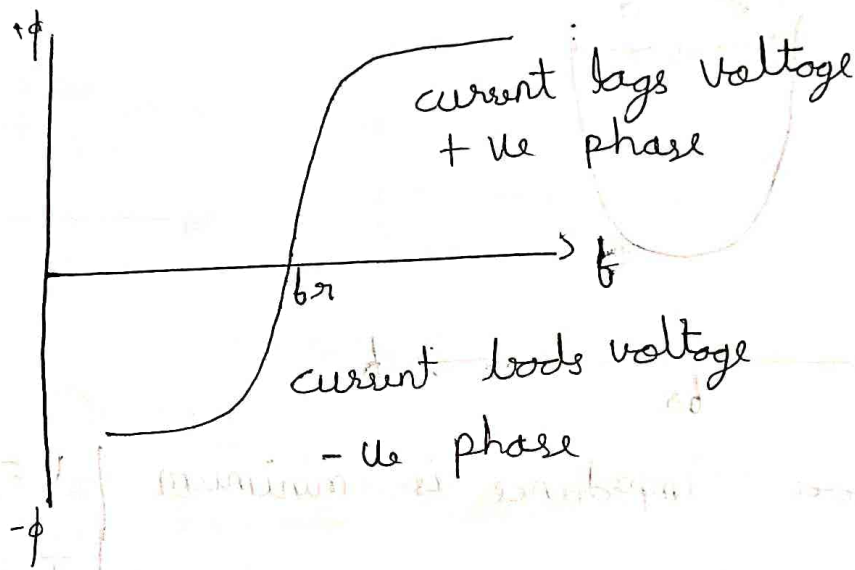
($X_C > X_L$) current leads the voltage.

$\therefore$ phase is ~~not~~ ~~positive~~ Negative.

• at frequencies above resonant frequency ($f > f_r$)

($X_L > X_C$) current lags the voltage

$\therefore$ phase is ~~not~~ positive

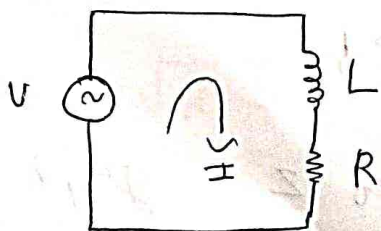


$\rightarrow$ Q factor (Quality factor)

It is given by the equation

$$Q = 2\pi \left(\frac{\text{maximum energy stored}}{\text{Total energy loss in a period of time}} \right)$$

$\rightarrow$ for Inductive circuit :



The maximum energy stored in the ckt is given by

$$\text{max energy} = I^2 \times L$$

$$\text{maximum energy} = \left(\frac{I_{ms}}{\sqrt{2}} \right)^2 \times L$$

$$\text{maximum energy} = \frac{I_m^2 L}{2} \dots \textcircled{1}$$

Total energy loss in the period is given by $P \times T$

$$\text{Total energy} = P \times T$$

$$= I^2 R \times T$$

$$= \frac{I_m^2}{2} R T \dots \textcircled{2}$$

①/② subs in

$$Q = 2\pi \frac{\frac{I_m^2 L}{2}}{\frac{I_m^2 R T}{2}}$$

$$Q = 2\pi \left(\frac{L}{R T} \right)$$

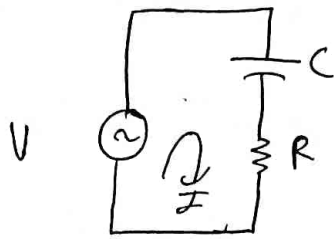
$$Q = \frac{2\pi f L}{R}$$

$$Q = \frac{\omega L}{R}$$

at resonance $\omega = \omega_0$

$$Q_0 = \frac{\omega_0 L}{R}$$

→ for capacitive circuit.



consider of Q of inductive ckt
at resonance we know that

$$\omega L = \frac{1}{\omega C}$$

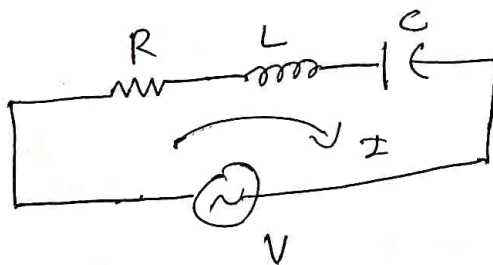
$$\omega_0 L = \frac{1}{\omega_0 C}$$

wkt

$$\therefore Q_0 = \frac{\omega_0 L}{R}$$

$$\therefore Q_0 = \frac{1}{\omega_0 C R}$$

→ Q factor for series RLC ckt



consider $Q = \frac{\omega_0 L}{R}$... ①

wkt

$$\omega_0 = 2\pi f_n$$

$$f_n = \frac{1}{2\pi\sqrt{LC}}$$

$$2\pi f_n = \frac{1}{\sqrt{LC}} \dots ②$$

sub ② in ①

$$Q = \frac{L}{\sqrt{LC} R}$$

$$Q = \frac{1}{\sqrt{C}} \frac{\sqrt{L}}{\sqrt{C}} \times \frac{1}{R}$$

$$Q = \left(\sqrt{\frac{L}{C}} \right) \frac{1}{R}$$

$$Q = \frac{1}{R} \left(\sqrt{\frac{L}{C}} \right)$$

→ series resonance ckt as voltage follower.

voltage across L & C under resonance

• For inductive circuit.

WKT

$$I_0 = \frac{V}{Z}$$

at resonance $Z = R$

$$I_0 = \frac{V}{R}$$

$$V_L = I_0 \times X_L$$

voltage across inductor

$$V_L = \frac{V}{R} j \omega L$$

$$V_L = \frac{V}{R} j \omega L$$

$$V_L = j \left(\frac{\omega L}{R} \right) V$$

WKT

$$\frac{V_L}{V} = Q_0$$

$$V_L = j Q_0 V$$

$$V_L = Q_0 V \angle 90^\circ$$

Q_0 = amplification factor

leads by 90°

For capacitive ckt:

$$I_0 = \frac{V}{Z}$$

at resonance.

$$I_0 = \frac{V}{R}$$

$$V_C = I_0(jX_C)$$

$$V_C = \frac{V}{R} - j \frac{1}{\omega C}$$

$$V_C = -j \left(\frac{1}{\omega C R} \right) V$$

We know that

$$\frac{1}{\omega C R} = Q_0$$

$$V_C = -j(Q_0) V$$

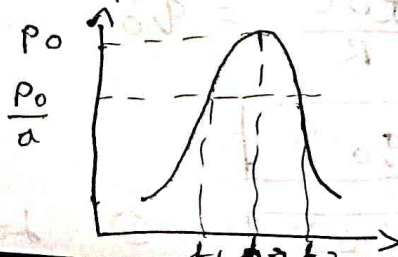
$$\boxed{V_C = Q_0 V \angle -90^\circ} \quad \text{② Voltage lags by } 90^\circ$$

at resonance, from the above exp it is understood that V_L & V_C have same magnitude but they are out of phase hence both gets cancelled at Resonance. The magnitude of V_L and V_C are Q_0 times the input voltage V , hence at resonance series RLC circuit works like voltage amplifier where Q_0 is the amplification factor.

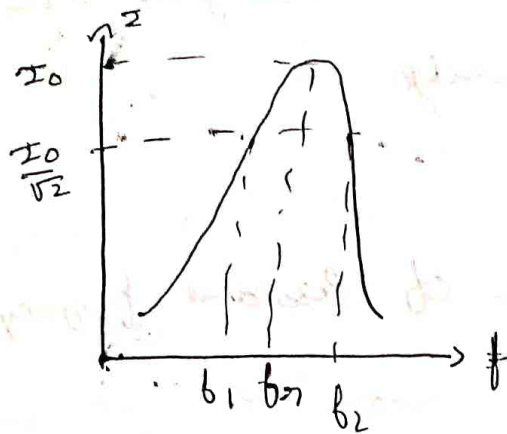
Bandwidth and selectivity of series resonance ckt.

Defined as Band of frequency the power in the ckt is half of its maximum value

$$BW = f_2 - f_1$$



In terms of current



general power equation:

$$P = I^2 R$$

The exp for maximum power is

$$P_{\text{max}} = I_{\text{max}}^2 R$$

WKT,

$$I_{\text{max}} = \frac{I_m}{\sqrt{2}}$$

$$P_{\text{avg}} = \frac{I_m^2}{2} R$$

$$P_{\text{avg}} = \frac{P_{\text{max}}}{2}$$

The frequency at which power in the ckt is half of its maximum value are called half power frequencies.

→ selectivity:

Defined as ability of ckt to distinguish b/w desired and undesired frequency

or

also defined as the Ratio of Resonance frequency to the BW of resonance ckt

$$S = \frac{f_r}{BW}$$

$$S = \frac{f_r}{b_2 - b_1}$$

$$S \propto \frac{1}{BW}$$

$$BW = \frac{R}{L}$$

$$BW = \frac{R}{2\pi L}$$

$$BW = b_2 - b_1$$

selectivity is given by.

$$S = \frac{f_r}{BW}$$

$$S = \frac{f_r}{\frac{R}{2\pi L}}$$

$$S = \frac{2\pi L f_r}{R}$$

$$S = \frac{\omega_0 L}{R}$$

$$S = Q_0$$

$$S < Q_0$$

$$Q_0 \propto \frac{1}{BW}$$

as Q_0 increases Bandwidth decreases and vice versa.

*** Relation ship b/w resonance & half power frequency.

or

Derive the exp. for half power / cutoff / 3dB frequency

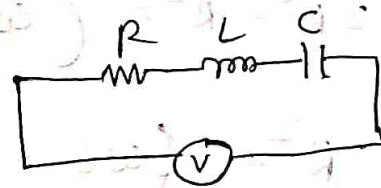
or

Exp for half power frequency in terms of BW etc.
in a series RLC ckt:

Let f_0 be the resonant frequency of series RLC ckt
and f_1 & f_2 be the half power frequency at
which current is $\frac{1}{\sqrt{2}}$ times of I_0

current in series RLC ckt is given by.

$$I = \frac{V}{|Z|}$$



we know that

$$Z = R + j(X_L - X_C)$$

$$|Z| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

$$\therefore I = \frac{V}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \quad \text{--- (1)}$$

current at half power frequency is given by

$$I = \frac{I_0}{\sqrt{2}}$$

$$I = \frac{1}{\sqrt{2}} \frac{V}{Z} \quad \text{--- (2)}$$

sub ② in ①

$$\frac{1}{\sqrt{2}} \frac{V}{|Z|} = \frac{V}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

at resonance

$$\frac{1}{\sqrt{2}} \cdot \frac{1}{R} = \frac{1}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

$$R\sqrt{2} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

square on both sides,

$$2R^2 = R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2$$

$$2R^2 - R^2 = \left(\omega L - \frac{1}{\omega C}\right)^2$$

$$R^2 = \left(\omega L - \frac{1}{\omega C}\right)^2$$

$$\boxed{\omega L - \frac{1}{\omega C} = \pm R} \dots \textcircled{3}$$

$$\text{Let } \boxed{\left(\omega_1 L - \frac{1}{\omega_1 C}\right) = -R} \dots \textcircled{4}$$

$$\boxed{\left(\omega_2 L - \frac{1}{\omega_2 C}\right) = +R} \dots \textcircled{5}$$

can ① adding 4 & ⑤

$$\left(\omega_1 L - \frac{1}{\omega_1 C}\right) + \left(\omega_2 L - \frac{1}{\omega_2 C}\right) = 0$$

$$\omega_1 L + \omega_2 L - \left(\frac{1}{\omega_1 C} + \frac{1}{\omega_2 C} \right) = 0$$

$$L(\omega_1 + \omega_2) = \frac{1}{C} \left(\frac{\omega_2 + \omega_1}{\omega_1 \omega_2} \right)$$

$$\therefore \boxed{\omega_1 \omega_2 = \frac{1}{LC}} \quad \dots \textcircled{6}$$

$$\text{WKT } \omega_0 = \frac{1}{\sqrt{LC}}$$

Square on both sides

$$\omega_0^2 = \frac{1}{LC} \quad \dots \textcircled{7}$$

from $\textcircled{6}$ & $\textcircled{7}$

$$\boxed{\omega_1 \omega_2 = \omega_0^2}$$

$$2\pi b_1 + 2\pi b_2 = (2\pi b_0)^2$$

$$\boxed{4\pi^2(b_1 b_2) = 4\pi^2 b_0^2}$$

$$b_0^2 = b_1 b_2$$

$$\boxed{b_0 = \sqrt{b_1 b_2}} \quad \dots \textcircled{A}$$

show that the geometric mean of two half power frequency is equal to b_0 (Resonating frequency).

case 2: subtracting eq (4) - (5)

$$\left(\omega_1 L - \frac{1}{\omega_1 C} \right) - \left(\omega_2 L - \frac{1}{\omega_2 C} \right) = -R - (+R)$$

~~by (-1)~~

$$(\omega_1 L - \omega_2 L) - \left(\frac{1}{\omega_1 C} + \frac{1}{\omega_2 C} \right) = -2R$$

$$L(\omega_1 - \omega_2) - \left(\frac{1}{\omega_1} + \frac{1}{\omega_2} \right) \frac{1}{C} = -2R$$

$$\frac{1}{C} \left(\frac{1}{\omega_1} - \frac{1}{\omega_2} \right) + (\omega_2 - \omega_1) L = 2R$$

$$(\omega_2 - \omega_1) L + \frac{1}{C} \left(\frac{\omega_2 - \omega_1}{\omega_1 \omega_2} \right) = 2R$$

$\div L$

$$(\omega_2 - \omega_1) + \frac{1}{LC} \left(\frac{\omega_2 - \omega_1}{\omega_1 \omega_2} \right) = \frac{2R}{L}$$

$$\frac{1}{LC} = \omega_1 \omega_2$$

$$(\omega_2 - \omega_1) + \cancel{\omega_1 \omega_2} \frac{(\omega_2 - \omega_1)}{\cancel{\omega_1 \omega_2}} = \frac{2R}{L}$$

$$2(\omega_2 - \omega_1) = \frac{R}{L}$$

$$2\pi f_2 - 2\pi f_1 = \frac{R}{L}$$

$$2\pi (f_2 - f_1) = \frac{R}{L}$$

$$b_2 - b_1 = \frac{R}{2\pi L}$$

$$BW = \frac{R}{2\pi L}$$

selectivity is given by.

$$S = \frac{b_2}{BW}$$

$$S = \frac{b_2}{\frac{R}{2\pi L}}$$

$$S = \frac{\omega_0 L}{R} = Q_0$$

$$S = Q_0$$

• Properties of series resonant circuit

- I_0 is in phase with V
- unity power factor
- at resonance impedance decreases current increases
- $I \uparrow \rightarrow \text{Power} \uparrow$
- Series RLC circuit works as voltage amplifier
- Resonance can be achieved.
 - By keeping b_1 constant varying L & C
 - By keeping L & C constant & varying F
- $Q_0 < 5$
- $BW < \frac{1}{S}$
- $Q_0 \approx \frac{1}{BW}$

→ Expression for ω_1 & ω_2

WKT

$$|Z| = \sqrt{2} R$$

$$\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} = \sqrt{2} R$$

squaring on both sides

$$\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}^2 = 2R^2$$

$$R^2 = \left(\omega L - \frac{1}{\omega C}\right)^2$$

$$\boxed{\omega L - \frac{1}{\omega C} = \pm R} \quad \dots \quad (1)$$

$$\omega_2 L - \frac{1}{\omega_2 C} = R$$

$$\frac{\omega_2^2 LC - 1}{\omega_2 C} = R$$

$$\omega_2^2 LC - 1 = R\omega_2 C$$

$$\omega_2^2 LC - \omega_2 RC - 1 = 0 \quad \text{--- (2)}$$

eq (2) is the form of quadratic eqn.

hence

$$ax^2 + bx + c = 0 \quad \text{--- (3)}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a = LC, \quad b = -RC, \quad c = -1$$

$$\omega_2 = \frac{-(-RC) \pm \sqrt{(-RC)^2 - (4)(LC)(-1)}}{2LC}$$

$$\omega_2 = \frac{RC \pm \sqrt{R^2 C^2 + 4LC}}{2LC}$$

$$\omega_2 = \frac{\frac{RC}{2LC} \pm \sqrt{\frac{R^2 C^2 + 4LC}{4L^2 C^2}}}{1}$$

$$\omega_2 = \frac{R}{2L} \pm \sqrt{\frac{R^2}{4L^2} + \frac{1}{LC}}$$

$$\omega_2 = \frac{R}{2L} \pm \sqrt{\frac{R^2}{4L^2} + \frac{1}{LC}} \quad \text{--- (7)}$$

put $\omega = 1$ in (1)

$$\omega_1 L - \frac{1}{\omega_1 C} = -R$$

$$\frac{\omega_1^2 LC - 1}{\omega_1 C} = -R$$

$$\omega_1^2 LC - 1 = -R\omega_1 C$$

$$\omega_1^2 LC + \omega_1 RC - 1 = 0 \quad (2)$$

2 is of the form aE

$$ax^2 + bx + c$$

$$a = LC \quad b = RC \quad c = -1$$

$$\omega_1 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\omega_1 = \frac{-RC \pm \sqrt{R^2 C^2 + 4LC}}{2LC}$$

$$\omega_1 = \frac{-R\cancel{C}}{2L\cancel{C}} \pm \sqrt{\frac{R^2\cancel{C}^2 + 4LC}{4L^2\cancel{C}^2}}$$

$$\omega_1 = -\frac{R}{2L} \pm \sqrt{\frac{R^2\cancel{C}^2}{4L^2\cancel{C}^2} + \frac{4LC}{4L^2\cancel{C}^2}}$$

$$\omega_1 = -\frac{R}{2L} \pm \sqrt{\frac{R^2}{4L^2} + \frac{1}{LC}}$$

$$\omega_1 = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \quad (4)$$

Expressing eqn ③ & ④ in terms of BW

$$BW = b_2 - b_1$$

$$BW = \omega_2 - \omega_1$$

$$BW = \left[\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \right] - \left[-\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \right]$$

$$\frac{R}{2L} + \cancel{\sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}} + \frac{R}{2L} - \cancel{\sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}}$$

$$BW = \frac{2R}{2L}$$

$$BW = \frac{R}{L}$$

to show that $b_0 = \sqrt{b_1 b_2}$ or prove that $b_0 =$ Geometric mean of two half power frequency / cutoff (3 db) frequency.

$$\omega_1 \omega_2 = \omega_0^2$$

$$\left[-\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \right] \cdot \left[\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \right] = \omega_0^2$$

$$= -\frac{R^2}{4L^2} - \frac{R}{2L} \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} + \frac{R}{2L} \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} + \frac{R^2}{4L^2} + \frac{1}{LC}$$

$$= -\frac{R^2}{4L^2} + \frac{R^2}{4L^2} + \frac{1}{LC}$$

$$= \frac{1}{LC} \dots \textcircled{5}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\omega_0^2 = \frac{1}{LC} \quad \text{.. (6)}$$

equating (5) & (6)

$$\omega_1 \omega_2 = \omega_0^2$$

$$\omega_0 = \sqrt{\omega_1 \omega_2}$$

$$2\pi f_0 = \sqrt{2\pi f_1 \cdot 2\pi f_2}$$

$$2\pi f_0 = \sqrt{4\pi^2 f_1 f_2}$$

$$2\pi f_0 = 2\pi \sqrt{f_1 f_2}$$

$$f_0 = \sqrt{f_1 f_2}$$

Prove that f_0 is the arithmetic mean of ω_1 & ω_2

W.K.T

$$\omega_1 = \frac{-R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} = \frac{-R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \omega_0^2}$$

$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} = \frac{R}{2L} + \omega_0 + \sqrt{\left(\frac{R}{2L}\right)^2}$$

If $\frac{R}{2L}$ is very much lesser than $\frac{1}{LC}$

$$\omega_1 = -\frac{R}{2L} + \sqrt{\frac{1}{LC}} = -\frac{R}{2L} + \omega_0$$

$$\omega_2 = -\frac{R}{2L} - \sqrt{\frac{1}{LC}} = -\frac{R}{2L} - \omega_0$$

$$\omega_1 + \omega_2$$

$$-\frac{R}{2L} + \omega_0 + \frac{R}{2L} - \omega_0$$

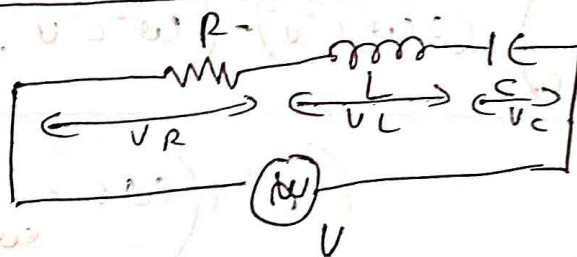
$$\omega_1 + \omega_2 = 2\omega_0$$

$$\omega_0 = \frac{\omega_1 + \omega_2}{2}$$

$$2\pi b_0 = 2\pi \frac{b_1 + b_2}{2}$$

$$b_0 = \frac{b_1 + b_2}{2}$$

express of ω (or ω) at which V_L is maximum.



$$V_L = IR$$

$$= IX_L$$

$$V_L = I\omega L \quad \dots (1)$$

Let

$$I = \frac{V}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}} \quad \dots (2)$$

$$\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}$$

Subs (2) in (1)

$$V_L = \omega L \frac{V}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \quad V_L^2 = \frac{\omega^2 L^2 V^2}{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

$$\frac{dV_L}{d\omega} = 0 \quad \text{for maximum voltage across inductor}$$

$$\frac{dV_L}{d\omega} = \frac{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2 \frac{d}{d\omega} \left(\omega^2 L^2 V^2 - \omega^2 L^2 V^2 \right) \frac{d}{d\omega} \left(\omega L - \frac{1}{\omega C} \right)}{\left(R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2 \right)^2} \quad - \left(\frac{1}{\omega^2} \right)$$

$$\frac{dV_L}{d\omega} = \frac{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2 \cdot 2\omega L^2 V^2 - \left(\omega^2 L^2 V^2\right) 2\left(\omega L - \frac{1}{\omega C}\right) \left(-\frac{1}{\omega^2}\right)}{\left(R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2\right)^2}$$

$$\frac{dV_L}{d\omega} = \frac{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2 \cdot (2\omega L^2 V^2) - \left(\omega^2 L^2 V^2\right) \left(2\left(\omega L - \frac{1}{\omega C}\right)\right)}{\left(R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2\right)^2}$$

$$0 = V^2 \left[R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2 (L^2 2\omega) - \omega^2 L^2 2\left(\omega L - \frac{1}{\omega C}\right) \left(L + \frac{1}{\omega^2 C}\right) \right]$$

$$\cancel{\left(R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2\right)} \cdot \cancel{2} L^2 \omega = \cancel{2} L^2 \omega^2 \left(\omega L - \frac{1}{\omega C}\right) \left(L + \frac{1}{\omega^2 C}\right)$$

$$R^2 + \left(\frac{\omega^2 LC - 1}{\omega C} \right)^2 = \omega^2 \left(\frac{\omega^2 LC + 1}{\omega^2 C} \right)$$

$$\frac{R^2 \omega^2 C^2 + (\omega^2 LC - 1)^2}{\cancel{\omega^2 C^2}} = \frac{\omega^3 LC + \omega}{\cancel{\omega^2 C^2}}$$

$$R^2 \omega^2 C^2 + (\omega^2 LC - 1)^2 = \omega^3 LC^2 + \omega C$$

$$R^2 \omega^2 C^2 + (\omega^2 LC - 1)^2 - \omega^3 LC^2 - \omega C = 0$$

$$R^2 \omega^2 C^2 + 2 - 2\omega^2 LC = 0$$

$$\omega^2 (R^2 C^2 - 2LC) = 0 \quad (1)$$

$$\omega^2 = \frac{-2}{R^2 C^2 - 2LC}$$

X -

$$\omega^2 = \frac{f R}{f (2LC - R^2 C^2)}$$

$$\boxed{\omega = \sqrt{\frac{2}{2LC - R^2 C^2}}}$$

$$\omega = \sqrt{\frac{2}{2LC \left(1 - \frac{R^2 C^2}{2L}\right)}}$$

$$\omega = \sqrt{\frac{1}{LC \left(1 - \frac{R^2 C}{2L}\right)}}$$

WKT $\frac{1}{\sqrt{LC}} = \omega_0$

$$\omega = \frac{\omega_0}{\sqrt{\left(1 - \frac{R^2 C}{2L}\right)}}$$

Parallel Resonance

Like series RLC ckt the resonance can be achieved in parallel RLC ckt, by varying frequency or LC components. In parallel RLC ckt the ckt behaves like a resistive network at Resonance. But resistance is maximum and current is minimum. Hence it is called as anti-resonance circuit.

- for parallel we need to consider admittance.

$$V|Y| = I$$

- parallel resonant ckt is analysed in terms of admittance Y

Note:

series $|Z| = \text{Reactance}$

parallel $|X| = \text{susptance} = B$

In case of admittance $B(L \& C)$ ckt have opposite signs

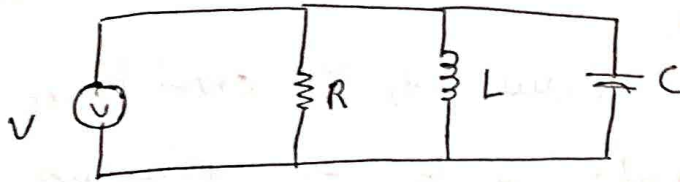
$$Y = \frac{1}{Z} = \frac{1}{R + jX} = G \pm jB$$

at particular frequency, under resonance, susceptance Inductance is equal to susceptance of capacitor and hence it gets cancelled.

$$Y = \frac{1}{R + jX}$$

$$Y = \frac{1}{R + j(X_L - X_C)}$$

Consider the following RLC ckt



The total impedance of the ckt

$$\frac{1}{Z} = \frac{1}{R} + \frac{1}{Z_L} + \frac{1}{Z_C}$$

$$Y = \frac{1}{R} + \frac{1}{jX_L} + \frac{1}{-jX_C}$$

$$= \frac{1}{R} - \frac{j}{\omega L} + \frac{j}{\omega C}$$

$$Y = \frac{1}{R} - \frac{j}{\omega L} + \frac{j}{\omega C}$$

$$Y = \frac{1}{R} - \frac{j}{\omega L} + j\omega C$$

$$Y = \frac{1}{R} + j \left(\omega C - \frac{1}{\omega L} \right) \quad \text{--- (1)}$$

at resonance

$$Y = \frac{1}{R}$$

or

$$Y = G$$

$G = \text{conductance}$

from (1)

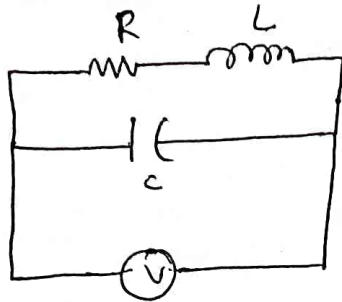
$$\therefore Y = G + jB$$

from eqn ① at resonance $X_L = X_C$

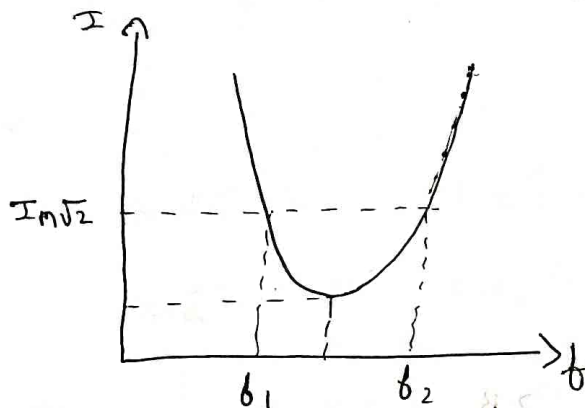
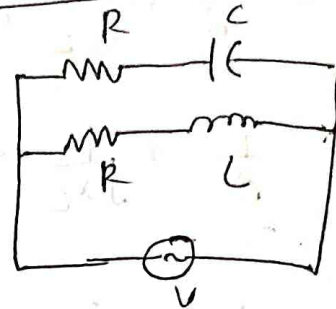
$$Y = \frac{1}{R}$$

which is the minimum value of Y and hence magnitude of Z increases which make I to be minimum.
(maximum)

→ case ②:



case ③:



→ Q factor:

defined as ratio of = $\frac{\text{current through capacitor}}{\text{total current}}$

$$Q = \frac{I_C}{I_R} = \frac{V/X_C}{V/R}$$

$$Q = \frac{R}{X_C} = \frac{2\pi f R C}{1}$$

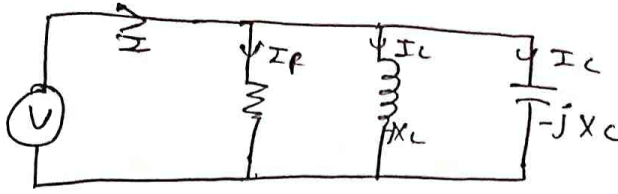
$$Q_C = 2\pi f R C$$

$$Q_C = \omega_0 R C$$

$$Q_L = \frac{R}{\omega_0 L}$$

Expression for ω_1 & ω_2 (when RLC are parallel)

consider 116 RLC ckt Case I



The Total current I is given by

$$I = I_R + I_L + I_C \quad \dots \quad (1)$$

$$\text{But } I = \frac{V}{R}$$

$$I = \frac{V}{R} + \frac{V}{jX_L} + \frac{V}{-jX_C}$$

$$I = \frac{V}{R} + \frac{V}{j\omega L} + \frac{V}{-j\omega C}$$

$$I = \frac{V}{R} \left[1 + j \left(\omega C - \frac{1}{\omega L} \right) \right]$$

$$I = V Y$$

where

$$Y = \frac{1}{R} + j \left(\omega C - \frac{1}{\omega L} \right)$$

$$|Y| = \sqrt{\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L} \right)^2} \quad \dots \quad (2)$$

$$|Y| = \frac{I}{V}$$

$$I = \frac{I_{\min} \sqrt{2}}{\sqrt{2}}$$

$$|Y| = \frac{I_{\min} \sqrt{2}}{V}$$

$$|Y| = \frac{\frac{V}{R} \sqrt{2}}{V}$$

$$|Y| = \frac{\sqrt{2}}{R} \dots (3)$$

equating (2) & (3)

$$\frac{\sqrt{2}}{R} = \sqrt{\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L} \right)^2}$$

squaring on both sides

$$\frac{2}{R^2} = \frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L} \right)^2$$

$$\frac{1}{R^2} = \left(\omega C - \frac{1}{\omega L} \right)^2$$

$$\left(\omega C - \frac{1}{\omega L} \right)^2 = \pm \frac{1}{R^2} \dots (4)$$

at $\omega = \omega_1$

$$\left(\omega_1 C - \frac{1}{\omega_1 L} \right) = -\frac{1}{R}$$

$$\frac{(\omega_1^2 C L - 1)}{\omega_1 L} = -\frac{1}{R}$$

$$R(\omega_1^2 C L - 1) = -\omega_1 L$$

$$\omega_1^2 RLC - R = -\omega_1 L$$

$$\omega_1^2 RLC + \omega_1 L - R = 0 \quad \therefore \quad \text{TE}$$

$$a = RLC = b = L \quad c = -R$$

$$x = \omega_1 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\omega_1 = \frac{-L \pm \sqrt{L^2 - 4(RLC)(-R)}}{2RLC}$$

$$\omega_1 = \frac{-L \pm \sqrt{L^2 + 4R^2LC}}{2RLC}$$

$$\omega_1 = \frac{-L}{2RLC} \pm \sqrt{\frac{L^2}{4R^2L^2C^2} + \frac{4R^2LC}{4R^2L^2C^2}}$$

$$\omega_1 = \frac{-1}{2RC} \pm \sqrt{\frac{1}{4R^2C^2} + \frac{1}{LC}}$$

$$\omega_1 = \frac{-1}{2RC} \pm \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}$$

$$\text{at } \omega = \omega_2$$

$$\omega_2 C - \frac{1}{\omega_2 L} = \frac{1}{R}$$

$$R(\omega_2^2 CL - 1) = \omega_2 L$$

$$\omega_2^2 RCL - R - \omega_2 L = 0$$

$$a = RCL \quad b = -R \quad c = -L$$

$$a = RLC \quad b = -L \quad c = -R$$

$$X = \omega_1 = \frac{L}{2RLC} \pm \frac{\sqrt{L^2 + 4R^2LC}}{2RLC}$$

$$X = \omega_2 = \frac{L}{2RLC} \pm \sqrt{\frac{L^2 + 4R^2LC}{4R^2L^2C^2}}$$

$$X = \omega_2 = \frac{1}{2RC} \pm \sqrt{\frac{L^2}{4R^2L^2C^2} + \frac{4R^2LC}{4R^2L^2C^2}}$$

$$\boxed{\omega_2 = \frac{1}{2RC} \pm \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}}$$

$$BW = \omega_2 - \omega_1$$

$$BW = \omega_2 - \omega_1$$

$$BW = \frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}} - \left[\frac{1}{2RC} - \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}} \right]$$

$$BW = \frac{1}{2RC} + \frac{1}{2RC}$$

$$BW = \frac{1}{2RC} (1 + 1)$$

$$BW = \frac{2}{2RC}$$

$$\boxed{BW = \frac{1}{RC}}$$

$$\boxed{BW = \frac{1}{RC 2\pi} \text{ Hz}}$$

→ Expressing ω_1 & ω_2 in terms of BW

$$\omega_1 = -\frac{BW}{2} + \sqrt{\left(\frac{BW}{2}\right)^2 + \frac{1}{LC}}$$

$$\omega_2 = \frac{BW}{2} + \sqrt{\left(\frac{BW}{2}\right)^2 + \frac{1}{LC}}$$

same that parallel resonant frequency is geometric mean of half power frequency.

$$\omega_1 \cdot \omega_2 = \left(-\frac{1}{2RC} + \sqrt{\frac{1}{2RC} + \frac{1}{LC}}\right) \cdot \left(\frac{1}{2RC} + \sqrt{\frac{1}{2RC} + \frac{1}{LC}}\right)$$

$$\omega_1 \cdot \omega_2 = -\frac{1}{4R^2C^2} - \left(\frac{1}{2RC} \left(\sqrt{\frac{1}{2RC} + \frac{1}{LC}}\right)\right) + \left(\frac{1}{2RC} \left(\sqrt{\frac{1}{2RC} + \frac{1}{LC}}\right)\right)$$

$$\left(\sqrt{\frac{1}{2RC} + \frac{1}{LC}}\right) \left(\sqrt{\frac{1}{2RC} + \frac{1}{LC}}\right)$$

$$\omega_1 \cdot \omega_2 = -\frac{1}{4R^2C^2} + \left(\sqrt{\frac{1}{2RC} + \frac{1}{LC}}\right)^2$$

$$\omega_1 \cdot \omega_2 = \cancel{-\frac{1}{4R^2C^2}} + \cancel{\frac{1}{2RC}} + \frac{1}{LC}$$

$$\omega_1 \cdot \omega_2 = \frac{1}{LC}$$

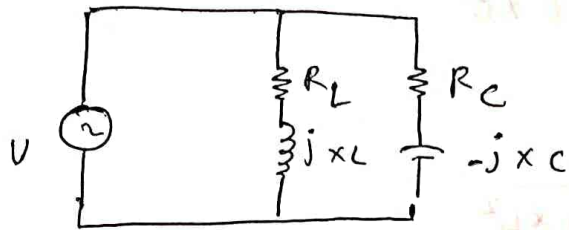
$$\omega_1 \cdot \omega_2 = \omega_0^2$$

$$\omega_0 = \sqrt{\omega_1 \cdot \omega_2}$$

$$\cancel{2\pi} b_0 = \cancel{2\pi} \sqrt{b_1 \cdot b_2}$$

$$b_0 = \sqrt{b_1 \cdot b_2}$$

Derive an Exp for the Resonant circuit also show that the circuit will resonate at all frequency if $R_L = R_C = \sqrt{\frac{L}{C}}$ where R_L is the Resistor in inductor branch, R_C is the Resistor in capacitor branch.



Consider parallel resonant ckt as shown above
wkt, $Y = G + jB$

The total admittance given by.

$$Y = Y_L + Y_C \quad \text{--- (2)}$$

$$Y_L = \frac{1}{Z_L} = \frac{1}{R_L + jX_L} \times \frac{R_L - jX_L}{R_L - jX_L}$$

$$Y_L = \frac{R_L - jX_L}{R_L^2 - (jX_L)^2} = \frac{R_L - jX_L}{R_L^2 + X_L^2} \quad \text{--- (3)}$$

$$Y_C = \frac{1}{Z_C} = \frac{1}{R_C - jX_C} \times \frac{R_C + jX_C}{R_C + jX_C}$$

$$Y_C = \frac{1}{Z_C} = \frac{R_C + jX_C}{R_C^2 + X_C^2} \quad \text{--- (4)}$$

subs (3) & (4) in (2)

$$Y = \frac{R_L - jX_L}{R_L^2 + X_L^2} + \frac{R_C + jX_C}{R_C^2 + X_C^2}$$

$$Y = \frac{R_L}{R_L^2 + X_L^2} - \frac{jX_L}{R_L^2 + X_L^2} + \frac{R_C}{R_C^2 + X_C^2} + \frac{jX_C}{R_C^2 + X_C^2}$$

$$Y = \frac{R_L}{R_L^2 + X_L^2} + \frac{R_C}{R_C^2 + X_C^2} + j \left(\frac{X_C}{R_C^2 + X_C^2} - \frac{X_L}{R_L^2 + X_L^2} \right)$$

comparing ⑤ & ①

$$G = \frac{R_L}{R_L^2 + X_L^2} + \frac{R_C}{R_C^2 + X_C^2}$$

$$B = \frac{X_C}{R_C^2 + X_C^2} - \frac{X_L}{R_L^2 + X_L^2}$$

at resonance $B_L = B_C$

$$B_L - B_C = 0$$

$$B = 0$$

$$\frac{X_C}{R_C^2 + X_C^2} - \frac{X_L}{R_L^2 + X_L^2} = 0$$

$$\frac{X_C}{R_C^2 + X_C^2} = \frac{X_L}{R_L^2 + X_L^2}$$

$$X_C \text{ as } \frac{1}{\omega C} \quad X_L = \omega L$$

$$\frac{1}{\omega (R_C^2 + \frac{1}{\omega^2 C^2})} = \frac{\omega L}{R_L^2 + X_L^2}$$

$$\cancel{\frac{R_L^2 + X_L^2}{\omega C}} = \cancel{(\omega^2 C^2 L)} (\omega^2 R_L C^3)$$

$$\cancel{R_L^2 + X_L^2} = \cancel{(\omega^2 C^2 L^2)} (\omega^2 R_L C^3)$$

$$\frac{\cancel{C} C}{R_L^2 \omega^2 C^2 + 1} = \frac{\cancel{L} L}{R_L^2 + \omega L^2}$$

$$R_L^2 C + \omega L^2 C = R_L^2 \omega^2 L C^2 + L$$

$$\omega^2 = \frac{L - C R_L^2}{L^2 C - L C^2 R_L^2}$$

$$\omega = \sqrt{\frac{L - C R_L^2}{L^2 C - L C^2 R_L^2}}$$

at Resonance

$$Y = \frac{R_L}{R_L^2 + X_L^2} + \frac{R_C}{R_C^2 + X_C^2}$$

$$X = \frac{R_L (R_C^2 + X_C^2) + R_C (R_L^2 + X_L^2)}{(R_L^2 + X_L^2) (R_C^2 + X_C^2)}$$

$$Y = \frac{R_L R_C^2 + R_L X_C^2 + R_C R_L^2 + R_C X_L^2}{R_L^2 R_C^2 + R_L^2 X_C^2 + R_C^2 X_L^2 + X_L^2 X_C^2}$$

Assume: $R_L = R_C = R$ $X_L^2 \times X_C^2 = R^4$

$$Y = \frac{R^3 + R X_C^2 + R^3 + R X_L^2}{R^4 + R^2 X_C^2 + R X_L^2 + \cancel{X_L^2 X_C^2} \rightarrow R^4}$$

$$Y = \frac{2R^3 + R X_C^2 + R X_L^2}{2R^4 + R^2 X_C^2 + R X_L^2}$$

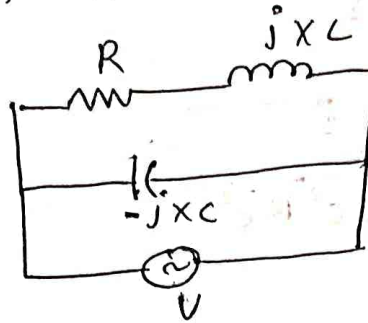
$$Y = \frac{R [2R^2 + X_C^2 + X_L^2]}{R^2 [2R^2 + X_C^2 + X_L^2]}$$

$$Y = \frac{1}{R}$$

$$Y = \frac{1}{R} = \sqrt{\frac{1}{4C}}$$

$$X = \sqrt{\frac{C}{L}}$$

Case 3: considers the network



$$Y = G + jB \quad \dots \quad (1)$$

The total admittance is given by

$$Y = Y_L + Y_C \quad \dots \quad (2)$$

$$Y_L = \frac{1}{Z_L} = \frac{1}{R + jX_C} \times \frac{R - jX_L}{R - jX_L}$$

$$Y_L = \frac{R - jX_L}{R^2 - (jX_L)^2} = \frac{R - jX_L}{R^2 + X_L^2} \quad \dots \quad (3)$$

$$Y_C = \frac{1}{Z_C} = \frac{1}{-jX_C} = \frac{j}{X_C} \quad \dots \quad (4)$$

Subs (3) & (4) in (2)

$$Y = \frac{R - jX_L}{R^2 + X_L^2} + \frac{j}{X_C}$$

$$Y = \frac{R}{R^2 + X_L^2} - \frac{jX_L}{R^2 + X_L^2} + \frac{j}{X_C}$$

$$Y = \frac{R}{R^2 + X_L^2} + j \left(\frac{1}{X_C} - \frac{X_L}{R^2 + X_L^2} \right) \quad \dots \quad (5)$$

comparing (5) & (1)

$$G = \frac{RL}{RL^2 + XL^2}$$

$$B = \frac{1}{\omega C} - \frac{XL}{RL^2 + XL^2}$$

at resonance $B = 0$

$$0 = \frac{1}{\omega C} - \frac{XL}{RL^2 + XL^2}$$

$$\frac{XL}{RL^2 + XL^2} = \frac{1}{\omega C}$$

$$XL \omega C = RL^2 + XL^2$$

$$XL \omega C - RL^2 - XL^2 = 0$$

$$\frac{L}{C} - RL^2 - (\omega L)^2 = 0$$

$$\frac{L}{C} - RL^2 - \omega^2 L^2 = 0$$

$$\frac{L}{C} - RL^2 = \omega^2 L^2$$

$$\omega^2 = \frac{\frac{L}{C} - RL^2}{L^2}$$

$$\omega^2 = \frac{L}{L^2 C} - \frac{RL^2}{L^2 C} = \frac{1}{LC} - \frac{R^2}{L C}$$

$$\omega = \sqrt{\frac{1}{LC} - \left(\frac{R_L}{L}\right)^2}$$

at resonant $B = 0$

$$\text{eq ① } Y = G_1 + jB$$

$$Y = G_1$$

$$Y = \frac{RL}{RL^2 + XL^2}$$

$$Y = \frac{RL}{RL^2 + \omega^2 L^2}$$

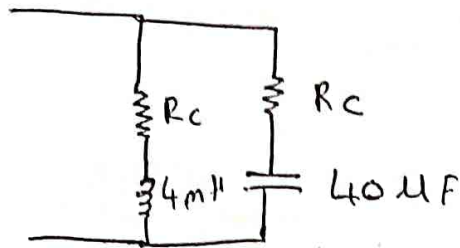
$$Y = \frac{RL}{RL^2 + \left(\sqrt{\frac{1}{LC} - \left(\frac{RL}{L}\right)^2}\right)^2 L^2}$$

$$Y = \frac{RL}{RL^2 + \left(\frac{1}{LC} - \left(\frac{RL}{L}\right)^2\right) L^2}$$

$$Y = \frac{RL}{\cancel{RL^2} + \frac{L}{C} - \cancel{RL^2}}$$

$$Y = \frac{RLC}{L}$$

→ Find the value of R_L and R_C for fig shown
 resonates at all frequency the ckt can resonate at
 any frequency if $R_L^2 = R_C^2 = \frac{L}{C}$

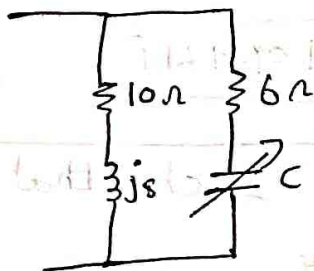


WKT $R_L = R_C = \sqrt{\frac{L}{C}}$

$$\sqrt{\frac{L}{C}} = \sqrt{\frac{4 \text{ m}}{40 \mu}} = 10 \Omega$$

$$\therefore R_L = R_C = 10 \Omega$$

→ Find the value of C for which ckt given in
 fig resonates at 750 Hz



$$Y = Y_L + Y_C$$

$$Y_L = \frac{1}{Z_L} = \frac{1}{10 + j8}$$

$$Y_C = \frac{1}{Z_C} = \frac{1}{6 - jX_C} = \frac{6 + jX_C}{6^2 - (jX_C)^2}$$

$$= 0.06 - 0.04j$$

$$= \frac{6 + jX_C}{36 + X_C^2}$$

$$Y_C = \frac{6 + jX_C}{36 + X_C^2}$$

$$y = \left(\frac{10}{164 + 36 + (x_c)^2} \right) + j \left(\frac{x_c}{36 + x_c^2} - \frac{8}{164} \right)$$

at resonance.

$$B = 0$$

$$y = \frac{x_c}{36 + x_c^2} = \frac{8}{164}$$

$$y = 164 x_c = 8 x_c^2 + 288$$

$$8 x_c^2 - 164 x_c + 288 = 0$$

$$x_{c1} = 18.56 \Omega$$

$$x_{c2} = 1.93 \Omega$$

$$C_1 = \frac{1}{\omega x_{c1}}$$

$$C_2 = \frac{1}{\omega x_{c2}}$$

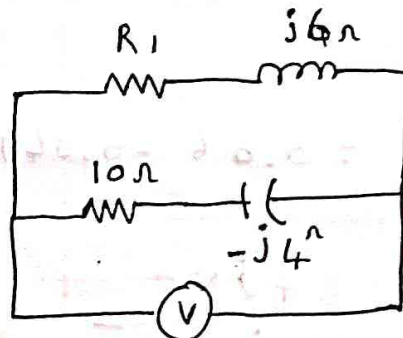
$$C_1 = \frac{1}{2\pi(750)18.56}$$

$$C_2 = \frac{1}{2\pi(750)1.93}$$

$$C_1 = 11.43 \mu F$$

$$C_2 = 109.9 \mu F$$

3) Determine the value of R_1 such that the ckt given in fig resonates



$$Y = Y_L + Y_C$$

$$Y_L = \frac{1}{Z_L} = \frac{1}{R_1 + j6} \quad \frac{R_1 - j6}{R_1^2 + 36}$$

$$Y_L = \frac{R_1 - j6}{R_1^2 + 36}$$

$$Y_C = \frac{1}{Z_C} = \frac{1}{10 - j4} \times \frac{10 + j4}{10 + j4}$$

$$Y_C = \frac{10 + j4}{100 + 16} \quad Y_C = \frac{10 + j4}{116}$$

$$\therefore Y = \left(\frac{R_1 - j6}{R_1^2 + 36} \right) + \left(\frac{10 + j4}{116} \right)$$

$$Y = G + jB$$

$$Y = \left(\frac{R_1}{R_1^2 + 36} + \frac{10}{116} \right) + j \left(\frac{4}{116} - \frac{6j}{R_1^2 + 36} \right)$$

at resonance $B = 0$

$$B = \frac{4}{116} - \frac{6}{R_1^2 + 36}$$

$$\frac{6}{R_1^2 + 36} = \frac{4}{116}$$

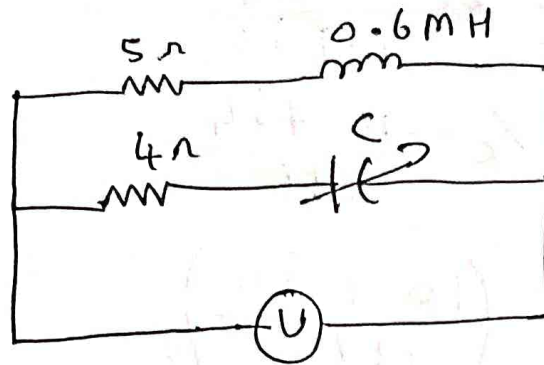
$$696 = 4R_1^2 + 144$$

$$R_1^2 = \frac{696 - 144}{4}$$

$$R_1 = \sqrt{138}$$

$$R_1 = 11.74 \Omega$$

→ for the ckt shown find two values of C when f of driving voltage is 5000 R/S



$$C_1 = 20.66 \mu\text{F}$$

$$C_2 = 121.06 \mu\text{F}$$

$$Y = Y_L + Y_C$$

$$Y_L = \frac{1}{Z_L} = \frac{1}{5 + j(0.6 \times 10^{-3})} \quad \frac{5 - j(0.6 \times 10^{-3})}{5 - j(0.6 \times 10^{-3})}$$

$$Y_L = \frac{5 - j(0.6 \times 10^{-3})}{25 + (0.36 \times 10^{-6})}$$

$$Y_C = \frac{1}{Z_C} = \frac{1}{4 + jX_C} \cdot \frac{4 - jX_C}{4 - jX_C} = \frac{4 - jX_C}{16 + (X_C)^2}$$

$$Y = \frac{5 - j(0.6 \times 10^{-3})}{25 + (0.36 \times 10^{-6})} + \frac{4 - jX_C}{16 + (X_C)^2}$$

$$Y = \left(\frac{5}{25 + 0.36 \times 10^{-6}} + \frac{4}{16 + (X_C)^2} \right) - j \left(\frac{0.6 \times 10^{-3}}{25.36} + \frac{X_C}{16 + X_C^2} \right)$$

at resonance $B = 0$

$$\frac{X_C}{16 + X_C^2} = \frac{0.6 \times 10^{-3}}{25 + 36 \times 10^{-6}}$$

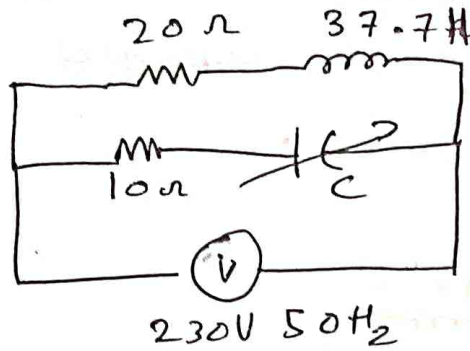
$$0.6 \times 10^{-3} X_C^2 - 25.36 X_C + 9.6 \times 10^{-3} = 0$$

$$X_{C2} = 0.02 - 3.9j$$

$$X_{C1} = 3.84 \times 10^{-4}$$

$$C_1 = \frac{1}{\omega X_{C1}} = \frac{1}{2\pi \times 5000 \times 3.84 \times 10^{-4}}$$

→ Find the value of C for circuit shown.



$$C_1 = 68.9 \mu F$$

$$C_2 = 1.46 mF$$

$$Y = Y_L + Y_C$$

$$Y_L = \frac{1}{20 + j(37.7)} \times \frac{20 - j(37.7)}{20 - j(37.7)} \quad Y_L = \frac{20 - j(37.7)}{400 + 1421.29}$$

$$Y_L = \frac{20 - j(37.7)}{400 + 1421.29}$$

$$Y_C = \frac{1}{10 - jX_C} \times \frac{10 + jX_C}{10 + jX_C} = \frac{10 + jX_C}{100 + X_C^2}$$

$$\therefore Y = \left(\frac{20}{1821.29} + \frac{10}{100 + X_C^2} \right) + j \left(\frac{X_C}{100 + X_C^2} - \frac{37.7}{1821.29} \right)$$

$$\therefore \frac{X_C}{100 + X_C^2} = \frac{37.7}{1821.29}$$

$$1821.29 X_C = 3771 + 37.71 X_C^2$$

$$-37.71 X_C^2 + 1821.29 X_C - 3771 = 0$$

$$X_{C1} = 46.129 \quad X_{C2} = 2.1678$$

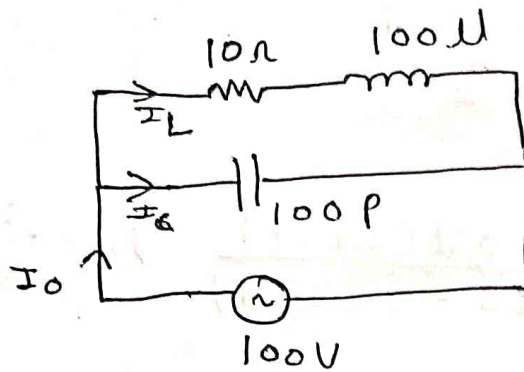
$$C_1 = \frac{1}{2\pi f X_{C1}}$$

$$C_2 = \frac{1}{2\pi f X_{C2}}$$

$$C_1 = 69.004 \mu F$$

$$C_2 = 1.46 mF$$

∴ For the llc resonant ckt shown find I_L , I_C , I_0 , and dynamic impedance for the ckt shown



$$Z_0 = \frac{L}{R_C}$$

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \left(\frac{R_L}{L}\right)^2}$$

$$f_0 = 1.591 \text{ MHz}$$

$$I_0 = \frac{V}{Z_0}$$

$$I_0 = \frac{100}{100 \text{ K}}$$

$$I_0 = 1 \text{ mA}$$

$$I_L = \frac{100}{R + jX_L}$$

$$I_L = \frac{100}{10 + j(2\pi f (100 \mu))}$$

$$I_L = 1.00 \times 10^{-3} - 0.1i$$

$$I_C = \frac{V}{X_C}$$

$$I_c = \frac{100}{\frac{1}{j\omega c}} = j100\omega c$$

$$I_c = j(100 \times 2\pi f \times 100 \times 10^{-12})$$

$$I_c = 0.099j$$

→ In a || RLC, $C = 50 \mu F$
define BW, Q, L and R

$$\omega_0 = \sqrt{\omega_1 \omega_2}$$

$$\omega_0^2 = \omega_1 \cdot \omega_2$$

$$\text{case (i)} \quad \omega_0 = 100, \quad \omega_2 = 120$$

$$\text{(ii)} \quad \omega_0 = 100, \quad \omega_1 = 80$$

$$\text{soln.} \quad \omega_0^2 = \omega_1 \cdot \omega_2$$

$$\omega_1 = \frac{\omega_0^2}{\omega_2} = \frac{100^2}{120} = 83.3 \text{ rad/s}$$

$$BW = \omega_2 - \omega_1$$

$$BW = 36.7 \text{ rad/s}$$

$$Q \text{ factor} = B.W = \frac{\omega_0}{\omega}$$

$$Q = \frac{\omega_0}{BW}$$

$$Q = 2.724$$

$$Q = \omega_0 R C$$

$$R = \frac{Q}{\omega_0 C} = 544 \Omega$$

$$Q = \frac{R}{\omega_0 L}$$

$$L = \frac{R}{\omega_0 Q}$$

$$\boxed{L = 2 \text{ H}}$$

$$\text{case ②: } \omega_0 = 100 \quad | \quad \omega_1 = 80$$

$$\omega_0^2 = \omega_1 \omega_2$$

$$\omega_2 = \frac{\omega_0^2}{\omega_1}$$

$$\omega_2 = 125 \text{ Rad/s}$$

$$BW = \omega_2 - \omega_1$$

$$BW = 125 - 80$$

$$\boxed{BW = 45 \text{ Rad/s}}$$

$$Q = \frac{\omega_0}{BW}$$

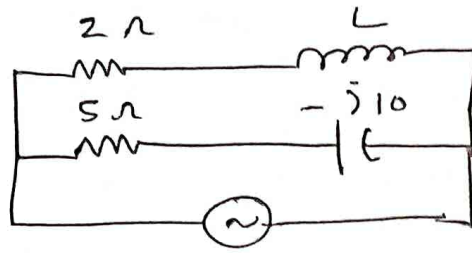
$$\boxed{Q = 2.22}$$

$$Q = \omega_0 RC$$

$$R = \frac{Q}{\omega_0 C} = 444 \Omega$$

$$L = \frac{R}{\omega_0 Q} = 2 \text{ H}$$

→ Find the value of ω which circuit resonates $\omega = 500 \text{ rad/sec}$



$$Y = Y_L + Y_C$$

$$Y_L = \frac{1}{Z_L} = \frac{1}{2 + jXL} = \frac{1}{2 + jXL} \cdot \frac{2 - jXL}{2 - jXL}$$

$$Y_L = \frac{2 - jXL}{4 + XL^2}$$

$$Y_C = \frac{1}{Z_C} = \frac{1}{5 - j10} = \frac{1}{5 - j10} \left(\frac{5 + j10}{5 + j10} \right)$$

$$Y_C = \frac{5 + j10}{25 + 100}$$

$$Y_C = \frac{5 + j10}{125}$$

$$Y = \frac{2 - jXL}{4 + XL^2} + \frac{5 + j10}{125}$$

$$Y = \frac{2}{4 + jXL^2} + \frac{5}{125} + j \left(\frac{10}{125} - \frac{XL}{4 + jXL^2} \right)$$

at resonance $B = 0$

$$\frac{10}{125} = \frac{XL}{4 + XL^2}$$

$$40 + 10XL^2 = 125XL$$

$$10XL^2 + 40 - 125XL = 0$$

$$XL_1 = 12.17$$

$$XL_2 = 0.3288$$

$$X_L = 2\pi f L$$

$$X_L = \omega L$$

$$L_1 = \frac{X_{L1}}{\omega} = \frac{12.17}{500} = 24.34 \text{ mH}$$

$$L_2 = \frac{X_{L2}}{\omega} = \frac{0.3286}{500} = 0.657 \text{ mH}$$